SYMPRAXIS CONSULTING

Building reusable code libraries for extending Microsoft 365

Julie Turner

SYMPRAXIS CONSULTING



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Julie Turner

Partner / CTO · Sympraxis Consulting

- Working with SharePoint since **2007**
- **Microsoft MVP**, Office Apps & Services since 2017
- M365 & Power Platform **Community Team** since 2019
- Open-source co-maintainer: **PnPjs** & **hTWOo**
- Co-host: ***Ask Sympraxis***

Agenda

1. Why Create a Code Library
2. Problem Statement
3. Architecture & Registries
4. Setup: Structure & Config
5. Development & Publishing
6. Security (.npmrc)
7. Live Demo: Build a Nav Menu

Why Create a Code Library

Reusable Code

- UX Components
- Services

Code Splitting

- Can be done with frameworks and bundlers



Problem Statement

SPFx Libraries

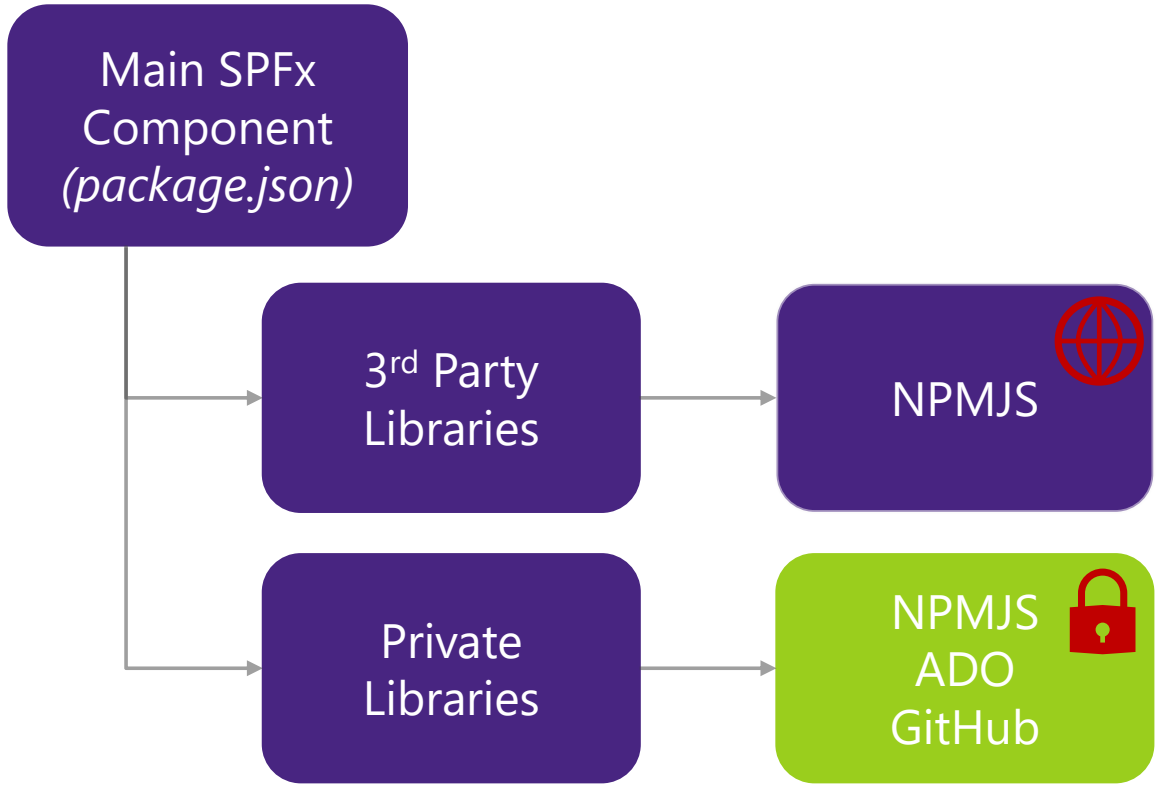
- npm link is finicky & painful (deleting the library folder)
- Only one version in app catalog at a time
- No complete control over the build chain
- Simple to get started

NPM Libraries

- Industry standard method of consuming libraries
- Supported by ADO, GitHub Enterprise, NPM (private)
- MANY ways to link and test
- Takes some time to setup
- Needs a delivery solution

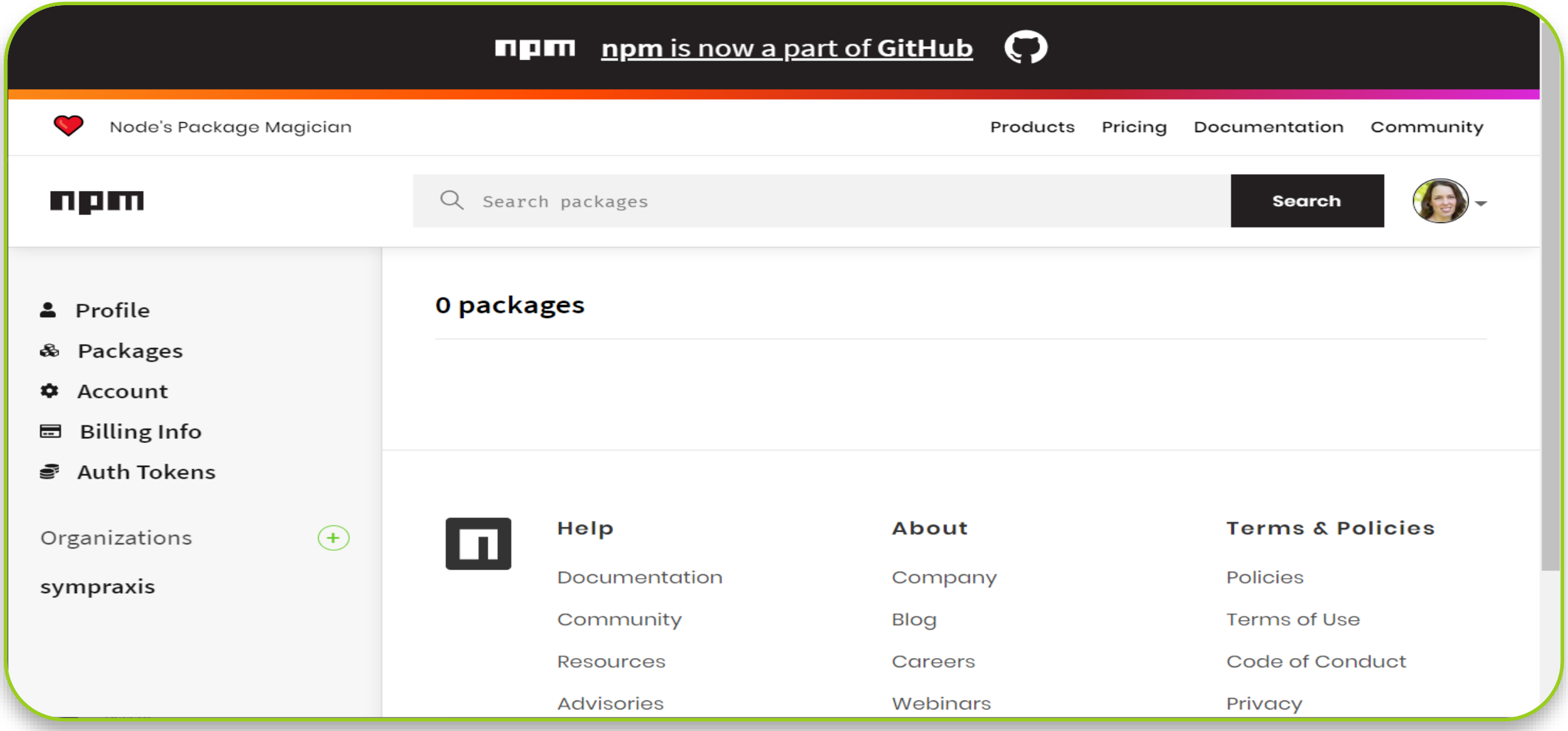
Main SPFx component includes private and public libraries
Public libraries come from NPMJS
Private libraries are built by organization and can be registered in GitHub, Azure DevOps, or NPMJS private subscription

Architecture

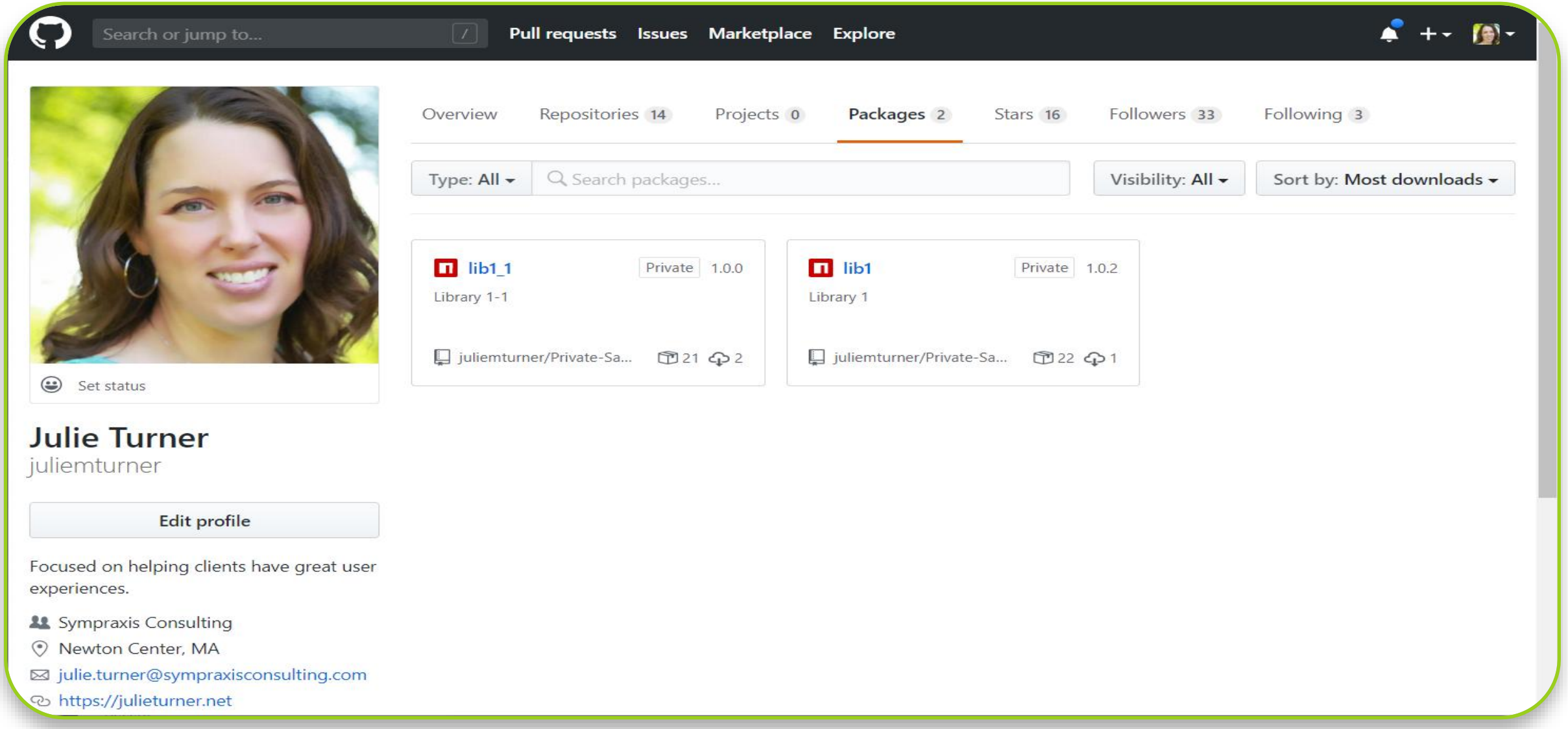


Architecture

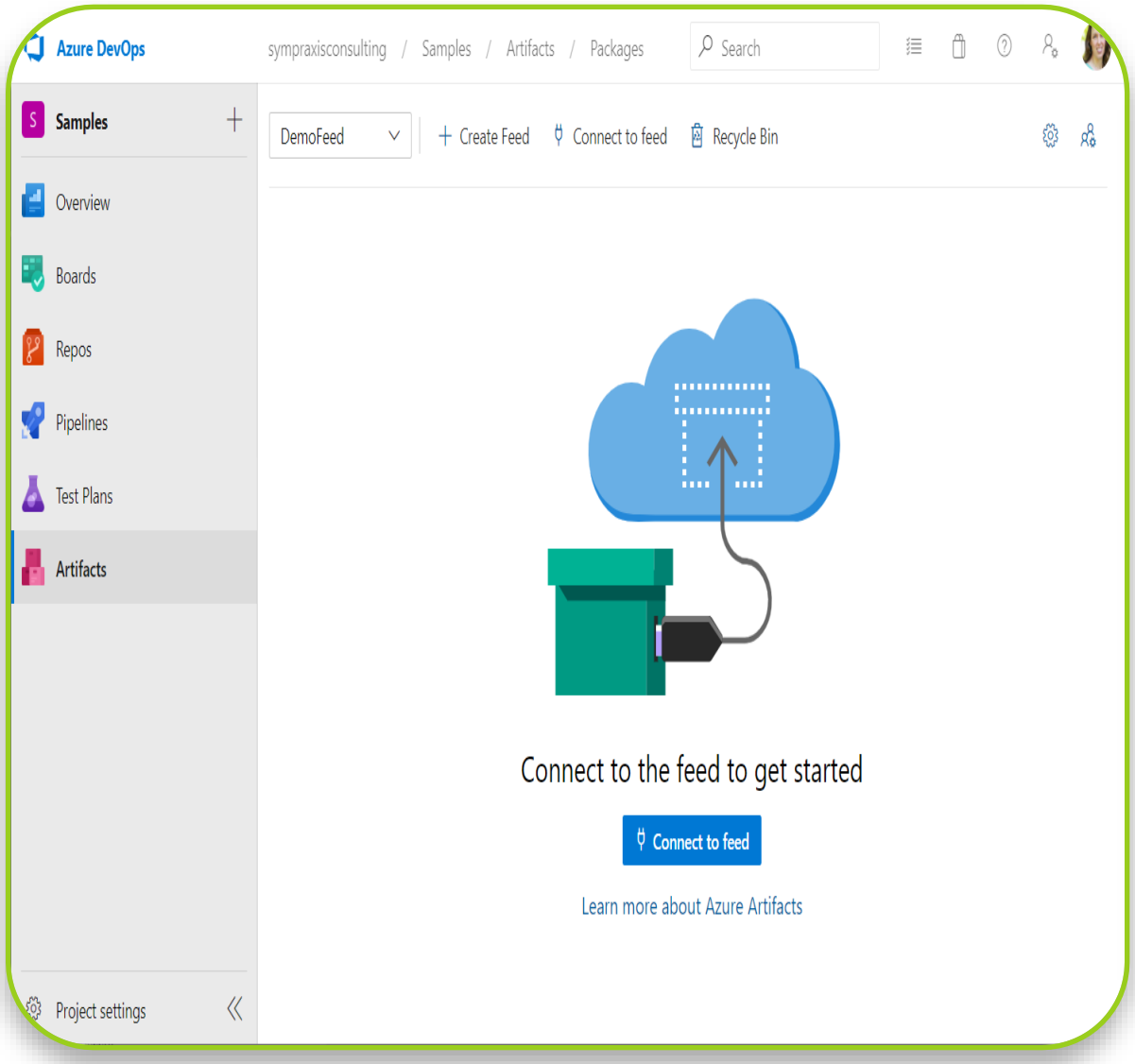
NPMJS

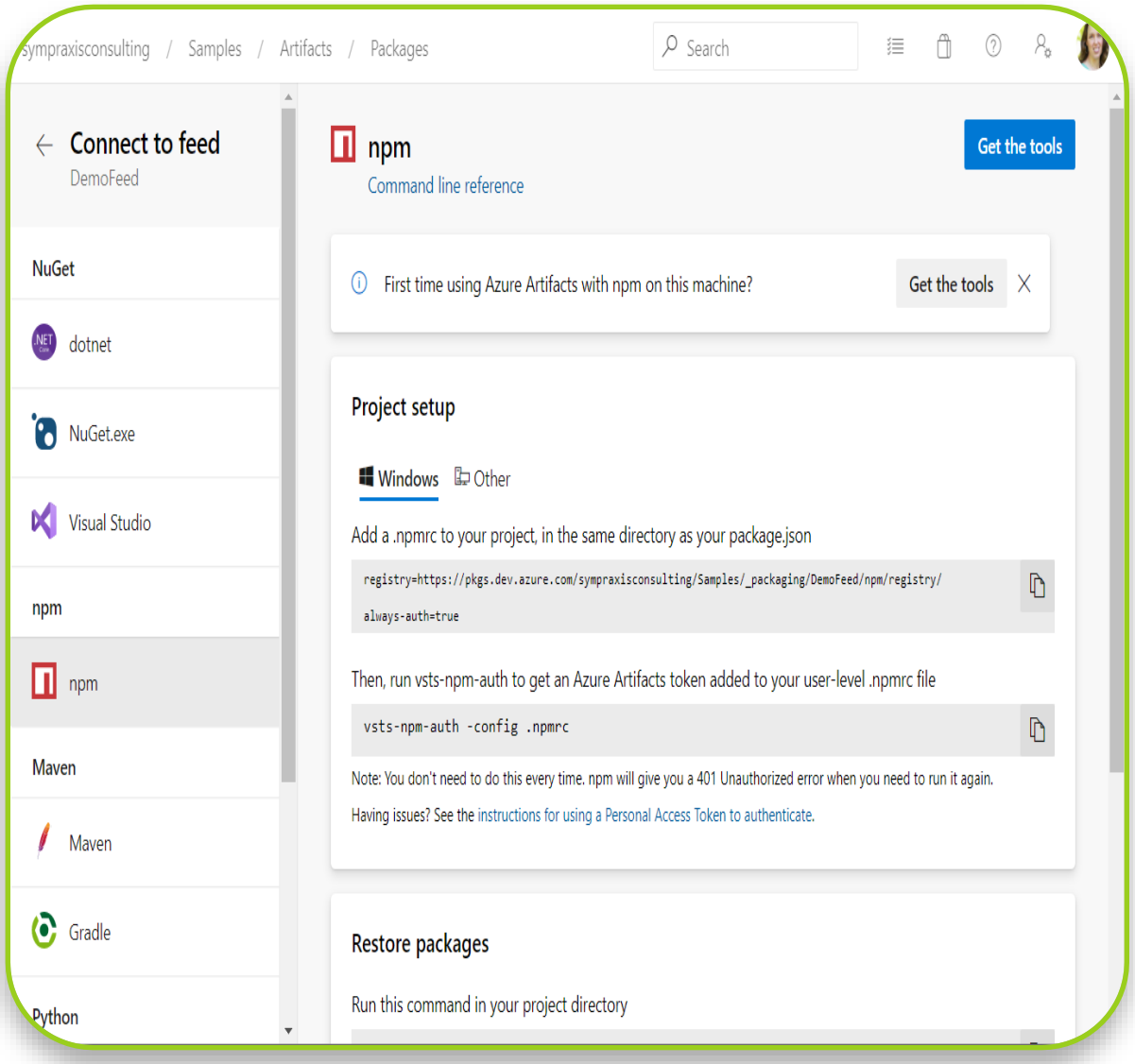

npm is now a part of GitHub
Node's Package Magician
Products
Pricing
Documentation
Community
npm
Search packages
Search
Profile
Packages
Account
Billing Info
Auth Tokens
Organizations
sympraxis
0 packages
Help
Documentation
Community
Resources
Advisories
About
Company
Blog
Careers
Webinars
Terms & Policies
Policies
Terms of Use
Code of Conduct
Privacy

GitHub Packages



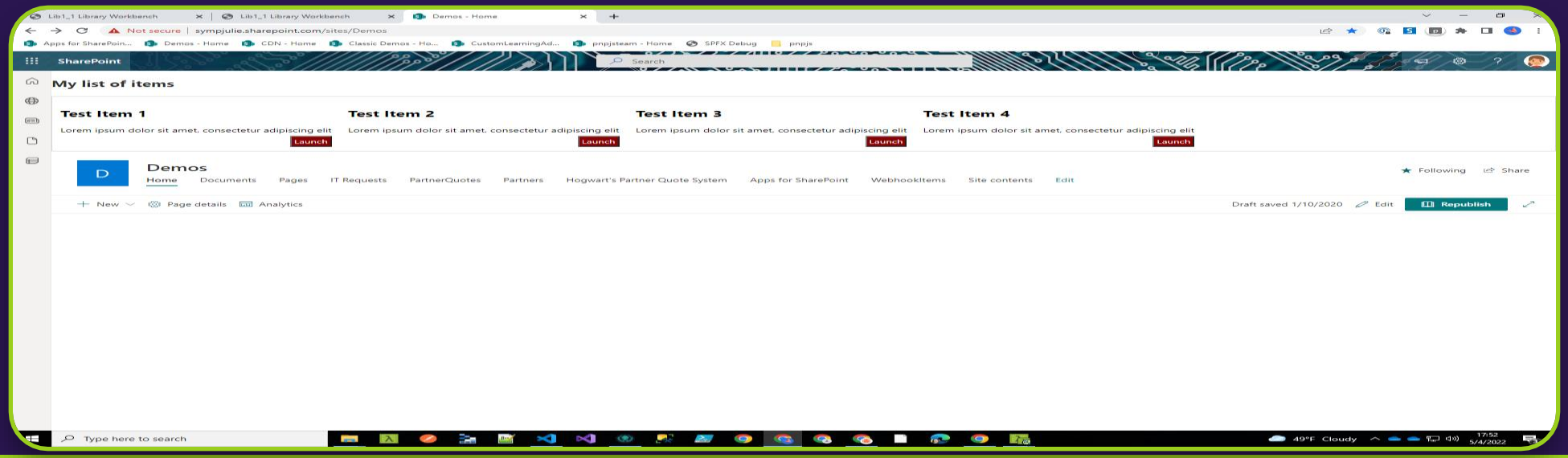
Azure DevOps Artifacts

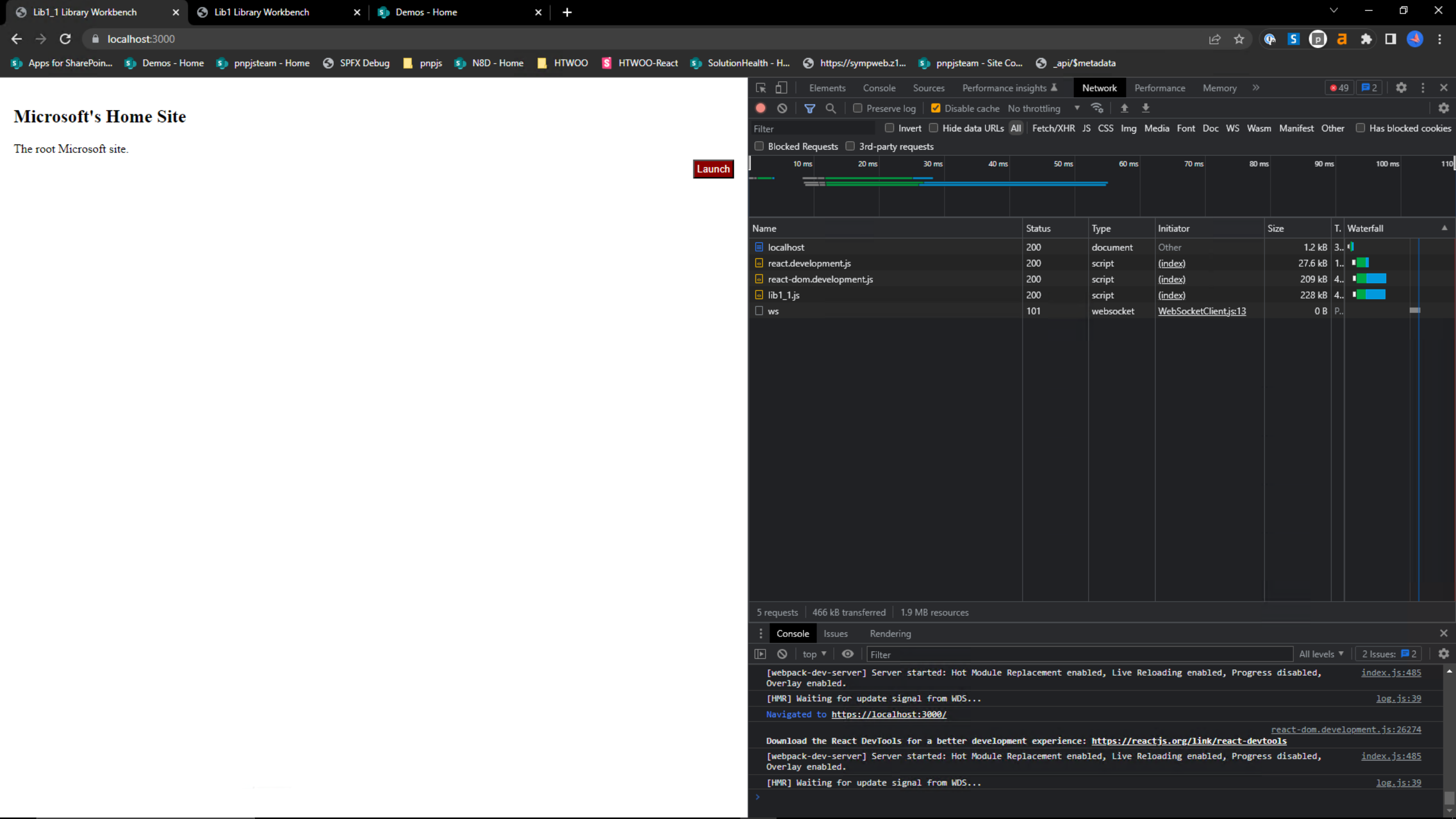


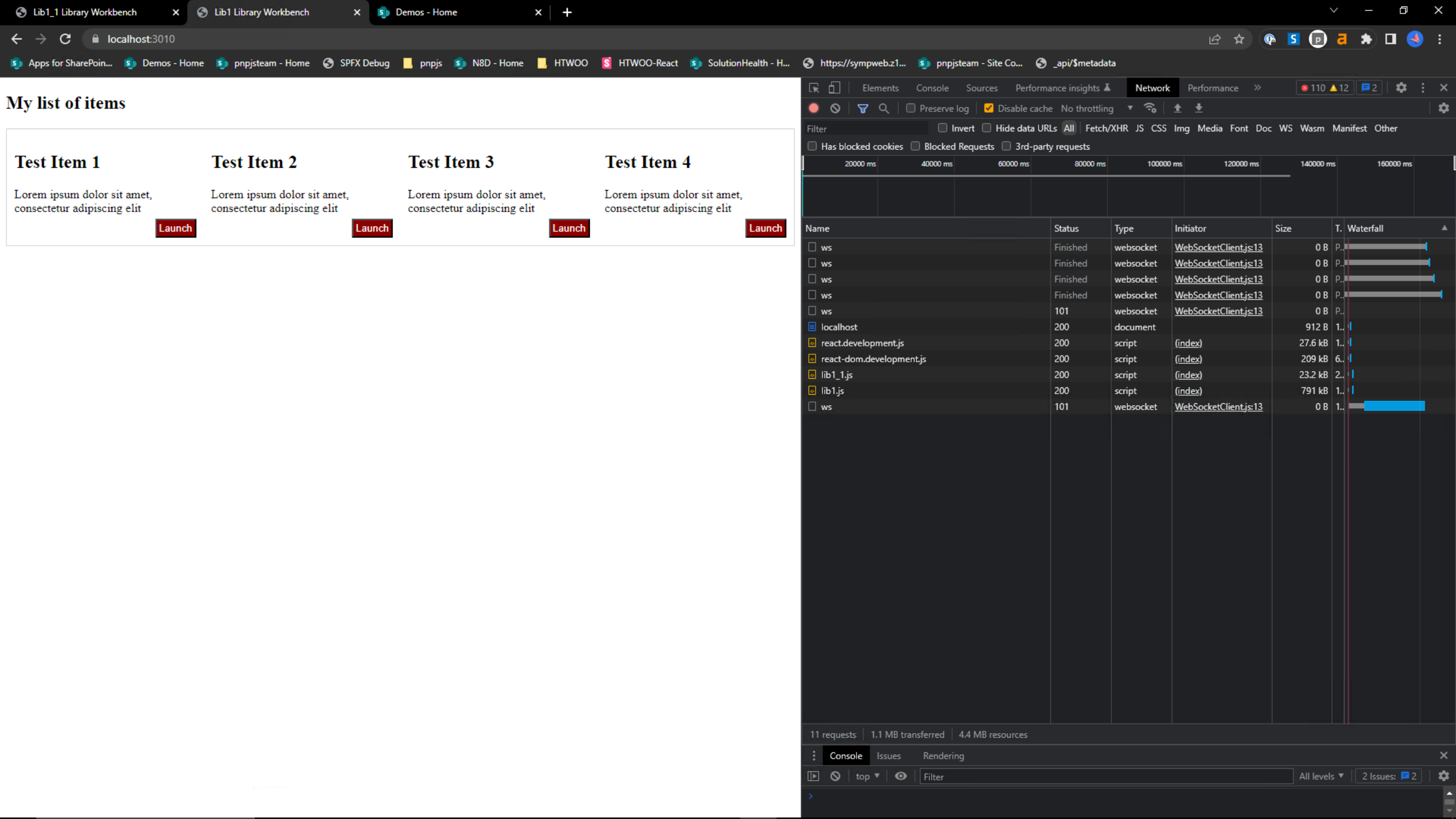


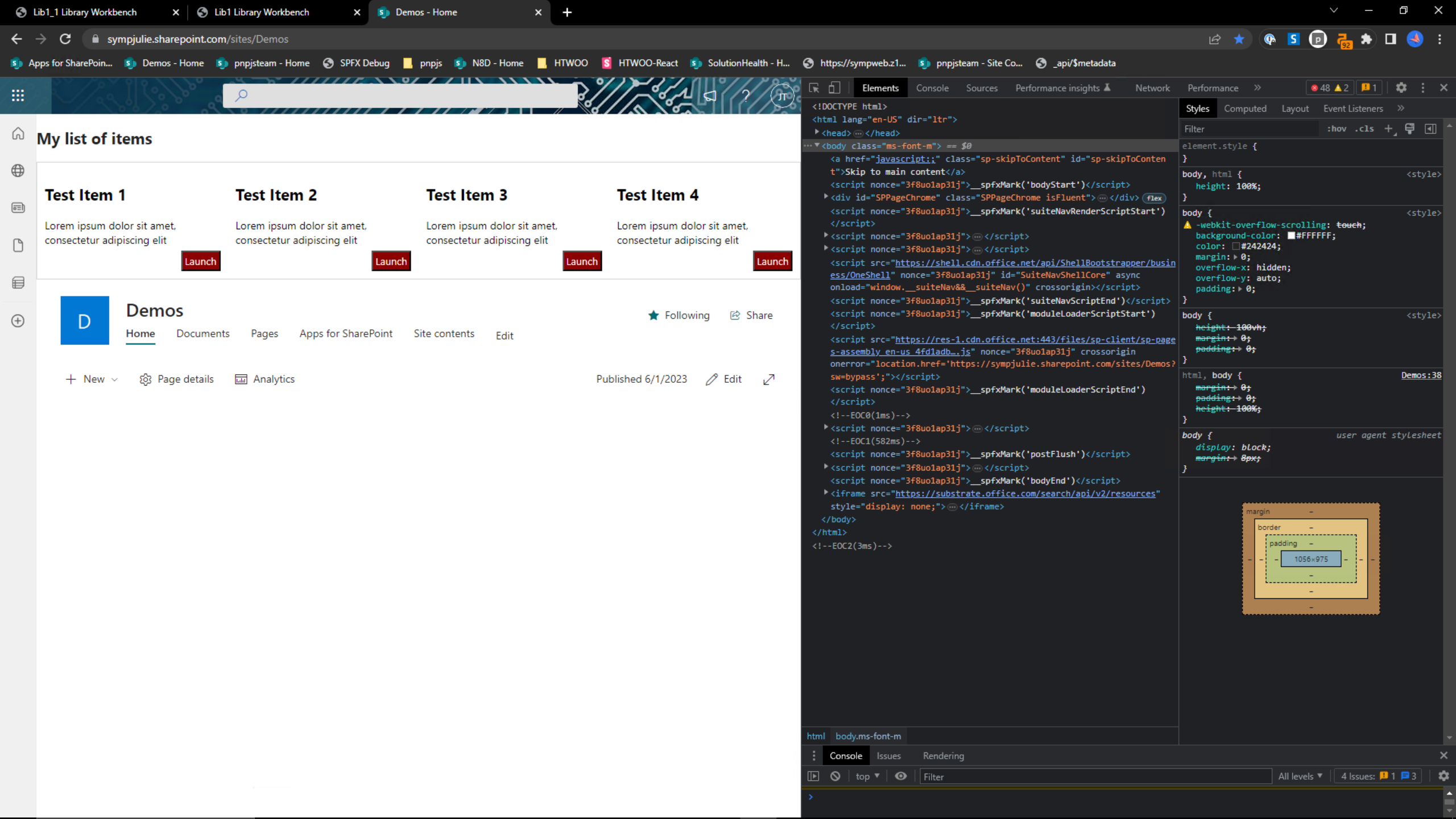
Let's build...

an ugly navigation menu




Lib1_1 Library Workbench
Lib1 Library Workbench
Demos - Home
localhost:3000
Apps for SharePoin...
Demos - Home
pnpjsteam - Home
SPFX Debug
pnpjs
N8D - Home
HTWOO
HTWOO-React
SolutionHealth - H...
https://sympweb.z1...
pnpjsteam - Site Co...
_api/$metadata
Microsoft's Home Site
The root Microsoft site.
Launch
Elements
Console
Sources
Performance insights
Network
Performance
Memory
Preserve log
Disable cache
No throttling
Filter
Invert
Hide data URLs
All
Fetch/XHR
JS
CSS
Img
Media
Font
Doc
WS
Wasm
Manifest
Other
Has blocked cookies
Blocked Requests
3rd-party requests
10 ms
20 ms
30 ms
40 ms
50 ms
60 ms
70 ms
80 ms
90 ms
100 ms
Name
Status
Type
Initiator
Size
T.
Waterfall
localhost 200 document Other 1.2 kB
react.development.js 200 script (index) 27.6 kB
react-dom.development.js 200 script (index) 209 kB
lib1_1.js 200 script (index) 228 kB
ws 101 websocket WebSocketClient.js:13 0 B
5 requests
466 kB transferred
1.9 MB resources
Console
Issues
Rendering
top
Filter
All levels
2 Issues: 2
[webpack-dev-server] Server started: Hot Module Replacement enabled, Live Reloading enabled, Progress disabled, Overlay enabled.
index.js:485
[HMR] Waiting for update signal from WDS...
log.js:39
Navigated to https://localhost:3000/
react-dom.development.js:26274
Download the React DevTools for a better development experience: https://reactjs.org/link/react-devtools
[webpack-dev-server] Server started: Hot Module Replacement enabled, Live Reloading enabled, Progress disabled, Overlay enabled.
index.js:485
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Lib1_1 Library Workbench
Lib1 Library Workbench
Demos - Home
localhost:3010
Apps for SharePoin...
Demos - Home
pnpjsteam - Home
SPFX Debug
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N8D - Home
HTWOO
HTWOO-React
SolutionHealth - H...
https://sympweb.z1...
pnpjsteam - Site Co...
_api/$metadata
My list of items
Test Item 1
Lorem ipsum dolor sit amet, consectetur adipiscing elit
Launch
Test Item 2
Lorem ipsum dolor sit amet, consectetur adipiscing elit
Launch
Test Item 3
Lorem ipsum dolor sit amet, consectetur adipiscing elit
Launch
Test Item 4
Lorem ipsum dolor sit amet, consectetur adipiscing elit
Launch
Elements
Console
Sources
Performance insights
Network
Performance
Preserve log
Disable cache
No throttling
Filter
Invert
Hide data URLs
All
Fetch/XHR
JS
CSS
Img
Media
Font
Doc
WS
Wasm
Manifest
Other
Has blocked cookies
Blocked Requests
3rd-party requests
20000 ms
40000 ms
60000 ms
80000 ms
100000 ms
120000 ms
140000 ms
160000 ms
Name
Status
Type
Initiator
Size
Waterfall
ws Finished websocket WebSocketClient.js:13 0 B
ws Finished websocket WebSocketClient.js:13 0 B
ws Finished websocket WebSocketClient.js:13 0 B
ws Finished websocket WebSocketClient.js:13 0 B
ws 101 websocket WebSocketClient.js:13 0 B
localhost 200 document 912 B
react.development.js 200 script (index) 27.6 kB
react-dom.development.js 200 script (index) 209 kB
lib1_1.js 200 script (index) 23.2 kB
lib1.js 200 script (index) 791 kB
ws 101 websocket WebSocketClient.js:13 0 B
11 requests
1.1 MB transferred
4.4 MB resources
Console
Issues
Rendering
top
Filter
All levels
2 Issues: 2


Lib1_1 Library Workbench
Lib1 Library Workbench
Demos - Home
sympjulie.sharepoint.com/sites/Demos
Apps for SharePoin...
Demos - Home
pnpjsteam - Home
SPFX Debug
pnpjs
N8D - Home
HTWOO
HTWOO-React
SolutionHealth - H...
https://sympweb.z1...
pnpjsteam - Site Co...
_api/$metadata
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Test Item 1
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Test Item 3
Lorem ipsum dolor sit amet, consectetur adipiscing elit
Launch
Test Item 4
Lorem ipsum dolor sit amet, consectetur adipiscing elit
Launch
D
Demos
Following
Share
Home
Documents
Pages
Apps for SharePoint
Site contents
Edit
New
Page details
Analytics
Published 6/1/2023
Edit
Elements
Console
Sources
Performance insights
Network
Performance
<!DOCTYPE html>
<html lang="en-US" dir="ltr">
<head></head>
<body class="ms-font-m"> == $0
Skip to main content
<script nonce="3f8uo1ap31j">__spfxMark('bodyStart')</script>

<script nonce="3f8uo1ap31j">__spfxMark('suiteNavRenderScriptStart')</script>
<script nonce="3f8uo1ap31j">__spfxMark('suiteNavScriptEnd')</script>
<script nonce="3f8uo1ap31j">__spfxMark('moduleLoaderScriptStart')</script>
<script nonce="3f8uo1ap31j">__spfxMark('moduleLoaderScriptEnd')</script>
<script nonce="3f8uo1ap31j">__spfxMark('postFlush')</script>
<script nonce="3f8uo1ap31j">__spfxMark('bodyEnd')</script>
<iframe src="https://substrate.office.com/search/api/v2/resources" style="display: none;"></iframe>
</body>
</html>
<!--EOC2(3ms)-->
Styles
Computed
Layout
Event Listeners
element.style {
}
body, html {
height: 100%;
}
body {
-webkit-overflow-scrolling: touch;
background-color: #FFFFFF;
color: #242424;
margin: 0;
overflow-x: hidden;
overflow-y: auto;
padding: 0;
}
body {
height: 100vh;
margin: 0;
padding: 0;
}
html, body {
margin: 0;
padding: 0;
height: 100%;
}
body {
display: block;
margin: 8px;
}
margin
border
padding
1056×975
html
body.ms-font-m
Console
Issues
Rendering
top
Filter
All levels
4 Issues: 1 3

Setup:

Code Structure

All code in one repository

- Folders in sibling hierarchy
- Single repo structure:
- myProject/
- ├─── SPFx Project
- ├─── Library 1
- └─── Library 2

Simplest setup — use NPM Workspaces (v7+, Node 15+) to link packages

Code in separate repositories

- Each library is its own repo
- Cloned in sibling hierarchy or anywhere on local machine
- Libraries linked as git submodules:
- myProject/ → myLibrary1/
- myProject/ → myLibrary2/

More isolation — good for libraries shared across multiple projects

NPM Workspaces (hybrid)

- Monorepo with workspace linking
- Other libraries added as submodules
- package.json at root declares workspaces
- NPM automatically links local packages
- Supported: NPM v7+, Node 15+

Best of both worlds — local dev speed with submodule isolation

Setup:

Bundled Reference

package.json – consumer

- Production: Install library as dev dependency from npm-style registry like any other library
- Development: Consider relative path install — npm i '../pathtolibrarypackage.json'

webpack.config.cjs – library

- Add `library` object to output property

Setup:

External Reference

package.json – consumer

- Install library as dev dependency from npm-style registry (or file reference)

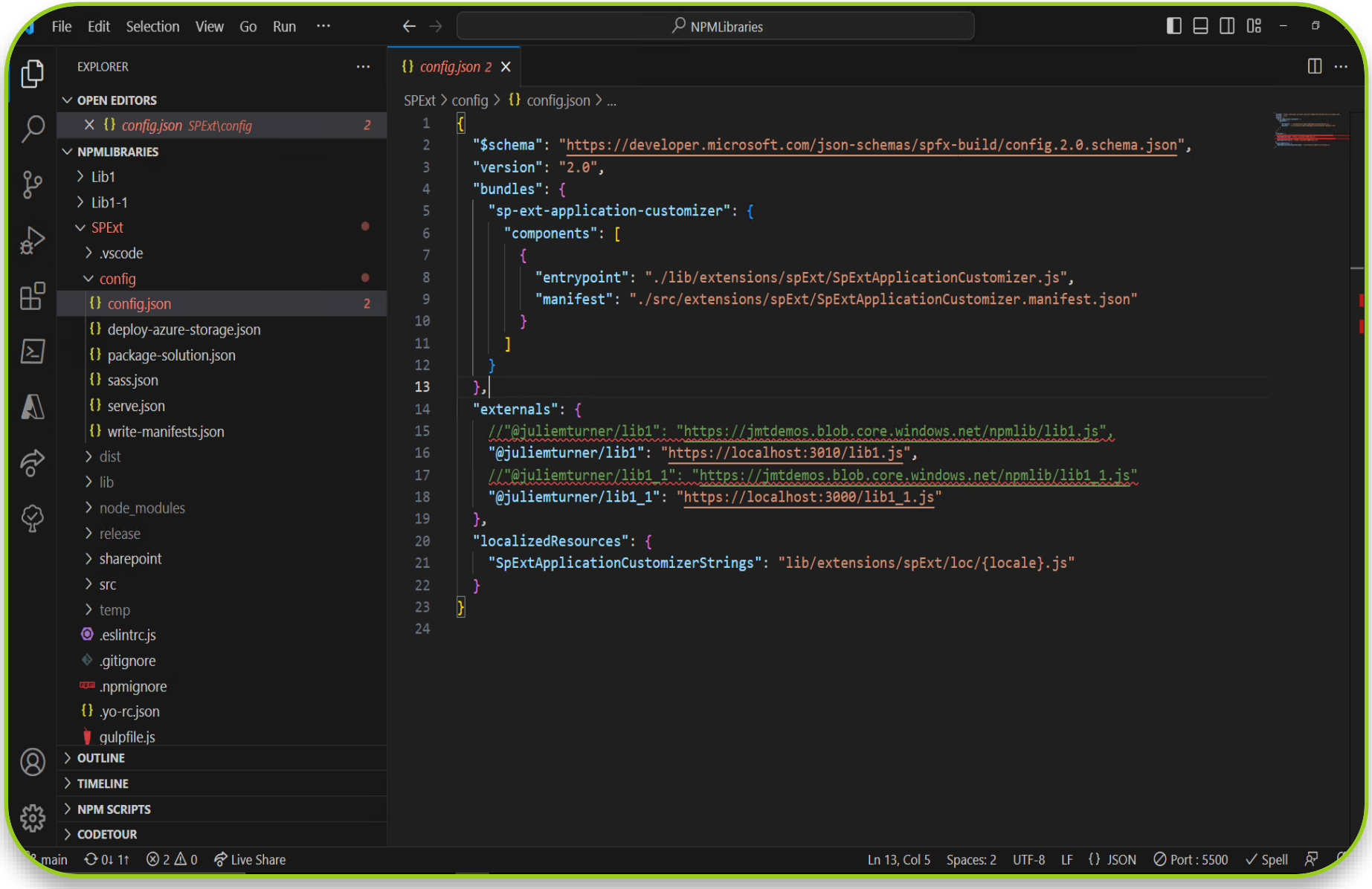
webpack.config.cjs – consumer

- List library in externals

webpack.config.cjs – library

- Add `library` object to output property

config.json



Setup:

SPFx

Config

Development & Production

Development of Library

- To test on its own, use webpack-dev-server
- To test bootstrapped into SPFx, modify config.json and do a development build

Using Library In Project

- Install from npm registry like any other library

NPM Publish

1

npm version first

Must run `npm version` before publishing — cannot reuse the same version

2

Credentials in .npmrc

Ensure .npmrc has credentials for your private registry

3

npm publish

Run `npm publish` — consider --dry-run to validate first

4

Production build first!

Ensure you've done a production build — use prepublish script in package.json

Security
.npmrc

```
# NPMJS Registry


registry=https://registry.npmjs.org/
//registry.npmjs.org/:_authToken=<YourNPMJSAuthToken>
```

```
# GitHub Registry

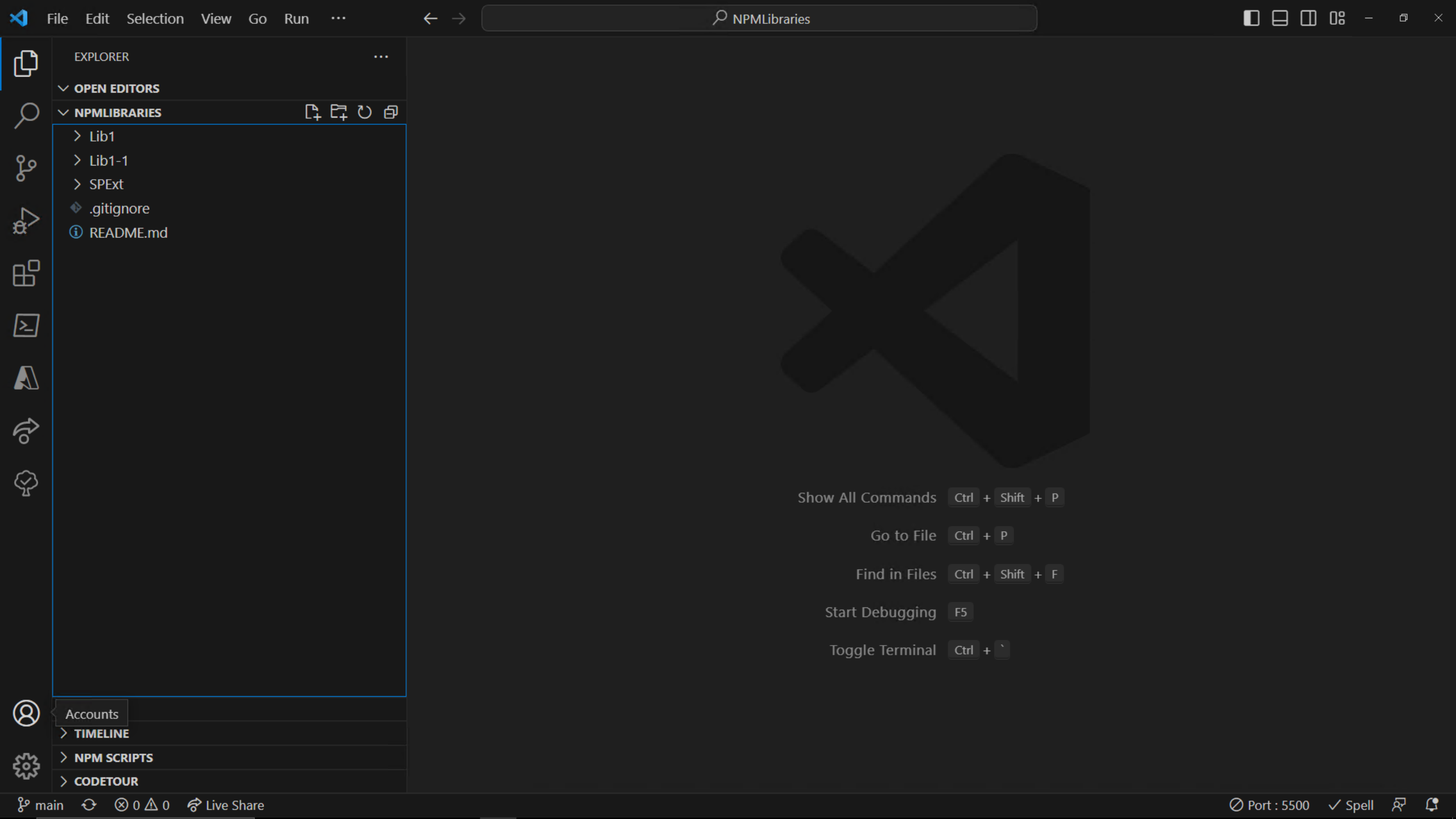

//npm.pkg.github.com/:_authToken=<YourPersonalAccessToken>
registry=https://npm.pkg.github.com/<YourGitHubAccount>/
@<YourGitHubAccount>:registry=https://npm.pkg.github.com
```

```
# Azure DevOps Registry


//pkgs.dev.azure.com/<Org>/_packaging/<Feed>/npm/registry/:_authToken=<Token>
@<Org>:registry=https://pkgs.dev.azure.com/<Org>/_packaging/<Feed>/npm/
//pkgs.dev.azure.com/<Org>/_packaging/<Feed>/npm/registry/:always-auth=true
```

Individual Menu Item

Lib1-1


File
Edit
Selection
View
Go
Run
NPMLibraries
EXPLORER
OPEN EDITORS
NPMLIBRARIES
Lib1
Lib1-1
SPExt
.gitignore
README.md
Show All Commands Ctrl + Shift + P
Go to File Ctrl + P
Find in Files Ctrl + Shift + F
Start Debugging F5
Toggle Terminal Ctrl + `
Accounts
TIMELINE
NPM SCRIPTS
CODETOUR
main
0 0
Live Share
Port : 5500
Spell

File Edit Selection View Go Run ··· NPMLibraries

EXPLORER

OPEN EDITORS
- package.json Lib1-1

NPMLIBRARIES
- .claude
- certs
- Lib1
- Lib1-1
  - app
  - node_modules
  - src
  - .eslintrc.js
  - package-lock-old.json
  - package-lock.json
  - package.json
  - tsconfig.json
  - webpack.config.js
- SPExt
- .gitignore
- README.md

Remote Explorer

OUTLINE
TIMELINE
AZURE: RESOURCES
AZURE: WORKSPACE
AZURE: HELP AND FEEDBACK
NPM SCRIPTS
CODETOUR

package.json

Lib1-1 > {} package.json > [ ] browserslist

```
{
  "name": "@juliemturner/lib1_1",
  "description": "Library 1-1",
  "license": "UNLICENSED",
  "version": "3.0.0",
  "private": false,
  "main": "./src/index.ts",
  "publishConfig": {
    "registry": "https://npm.pkg.github.com/juliemturner/"
  },
  "repository": {
    "type": "git",
    "url": "ssh://git@github.com/juliemturner/Private-Samples.git"
  },
  "engines": {
    "node": ">=0.10.0"
  },
  Debug
  "scripts": {
    "build": "run-s clean package",
    "package": "webpack",
    "watch": "webpack --watch",
    "start": "webpack-dev-server --config ./webpack.config.js",
    "clean": "del-cli \"./?(dist|lib)\"",
    "lint": "eslint --ext \"./src/**/*.{ts,tsx}\""
  },
  "browserslist": [
    "last 1 version",
    "> 1%",
    "maintained node versions",
    "not dead"
  ],
  "devDependencies": {
```

main 0 0 Live Share [Azurite Table Service] [Azurite Queue Service] [Azurite Blob Service] Julie Turner (3 years ago) Ln 26, Col 20 Spaces: 2 UTF-8 CRLF { } JSON Go Live

Lib1-1

EXPLORER

OPEN EDITORS
- package.json Lib1-1

NPMLIBRARIES
- .claude
- certs
- Lib1
- Lib1-1
  - app
  - node_modules
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  - package-lock.json
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- README.md

Remote Explorer

OUTLINE
TIMELINE
AZURE: RESOURCES
AZURE: WORKSPACE
AZURE: HELP AND FEEDBACK
NPM SCRIPTS
CODETOUR

package.json

Lib1-1 > {} package.json > { } engines

```
26    "browserslist": [
27      "last 1 version",
28      "> 1%",
29      "maintained node versions",
30      "not dead"
31    ],
32    "devDependencies": {
33      "react": "17.0.1",
34      "react-dom": "17.0.1",
35      "@types/react": "17.0.45",
36      "@types/webpack": "^5.28.1",
37      "@typescript-eslint/eslint-plugin": "^5.59.8",
38      "@typescript-eslint/parser": "^5.58.8",
39      "css-loader": "^6.8.1",
40      "del-cli": "^5.0.0",
41      "eslint": "^8.41.0",
42      "npm-run-all": "^4.1.5",
43      "sass": "^1.62.1",
44      "sass-loader": "^13.3.1",
45      "style-loader": "^3.3.3",
46      "ts-loader": "^9.5.1",
47      "typescript": "^5.4.5",
48      "webpack": "5.95.0",
49      "webpack-cli": "^5.1.1",
50      "webpack-dev-server": "^4.15.0"
51    },
52    "files": [
53      "/src",
54      "/dist"
55    ]
56  }
57
```

Lib1-1

Source Control (Ctrl+Shift+G)

webpack.config.js


```
1   const path = require('path');
2   const fs = require('fs');
3
4   module.exports = (env, argv) => {
5     return {
6       mode: (env.WEBPACK_SERVE || env.WEBPACK_WATCH) ? 'development' : 'production',
7       target: 'web',
8       entry: {
9         lib1_1: path.resolve(__dirname, './src/index.ts')
10      },
11      output: {
12        path: path.resolve(__dirname, 'dist'),
13        filename: "[name].js",
14        library: { type: "umd", name: "lib11" },
15        clean: true
16      },
17      devtool: 'inline-source-map',
18      resolve: {
19        extensions: ['.ts', '.tsx', '.js', '.scss'],
20        modules: ['node_modules']
21      },
22      module: {···
45      },
46      externals: {
47        'react': { root: 'React', commonjs: 'react', commonjs2: 'react', amd: 'react' },
48        'react-dom': { root: 'ReactDOM', commonjs: 'react-dom', commonjs2: 'react-dom', amd: 'react-dom' }
49      },
50      devServer: {···
71      }
72    };
73  };
74
75
```

Lib1-1

webpack.config.js

Lib1-1 > webpack.config.js > ...

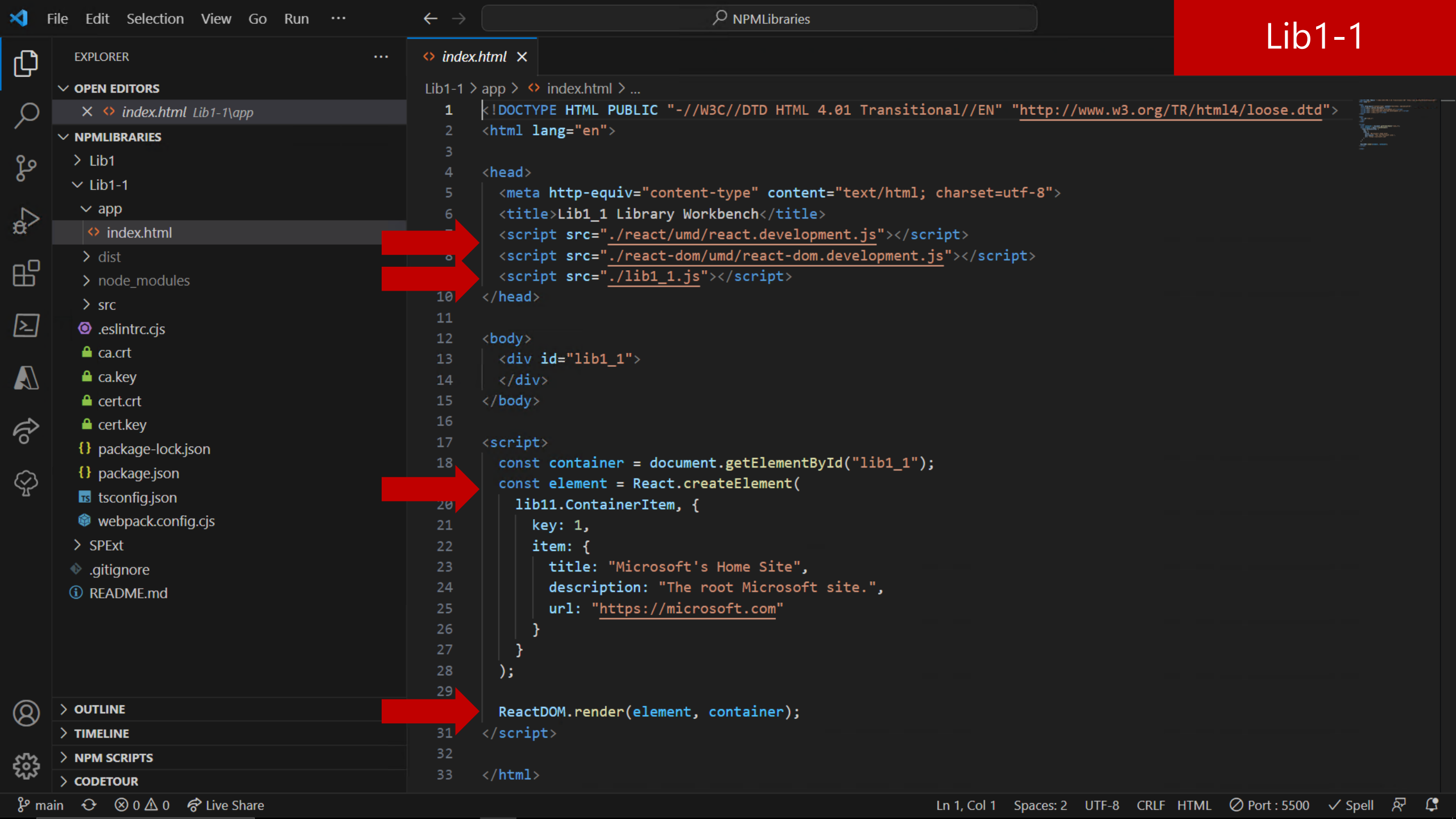
```
4   module.exports = (env, argv) => {
8       entry: {
10      },
11      output: {···
16      },
17      devtool: 'inline-source-map',
18      resolve: {···
21      },
        module: {
23        rules: [{
24          test: /\.tsx|.ts?$/,
25          exclude: /node_modules/,
26          use: [
27            {
28              loader: 'ts-loader',
29              options: {
30                happyPackMode: true,
31                transpileOnly: true,
32              },
33            },
34          ]
35        },
36        {
37          test: /\.s[ac]ss$/i,
38          exclude: /node_modules/,
39          use: ["style-loader", "css-loader", { loader: "sass-loader", options: { api: "modern" } }]
40        },
41        {
42          test: /\.svg/,
43          type: 'asset/resource'
44        }]
45      },
46      externals: {
47        'react': { root: 'React', commonjs: 'react', commonjs2: 'react', amd: 'react' },
```

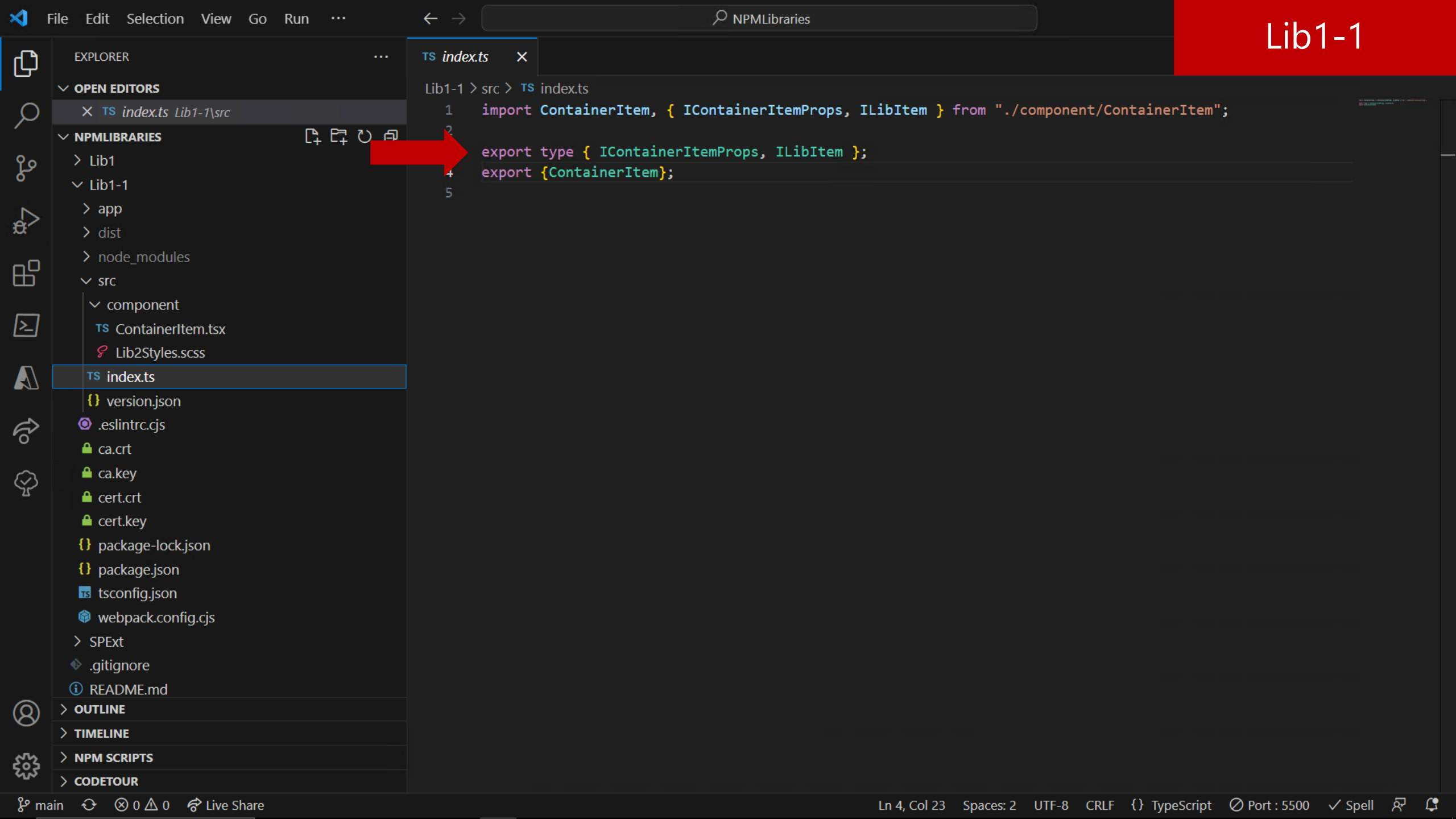
Lib1-1

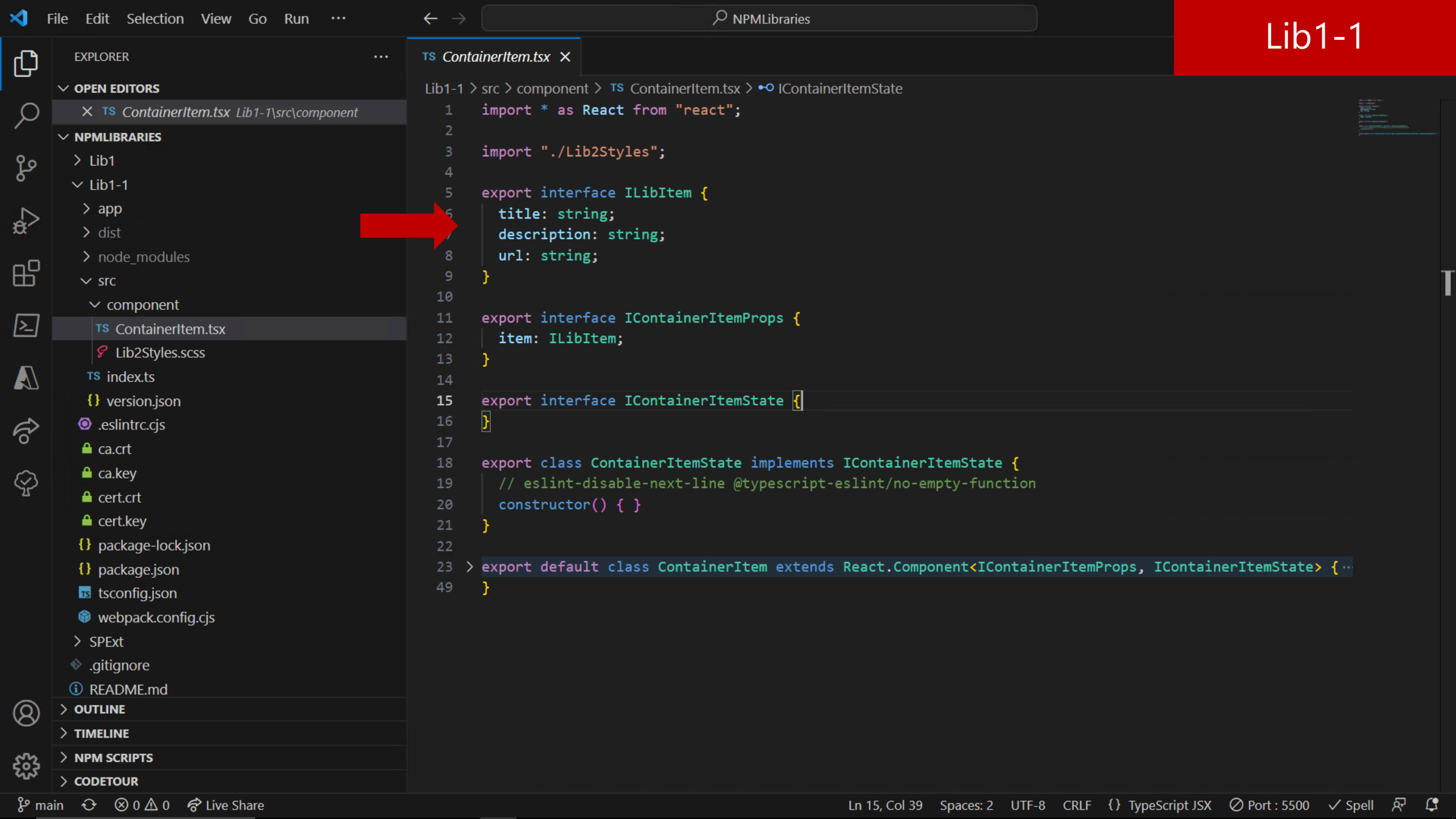
webpack.config.js

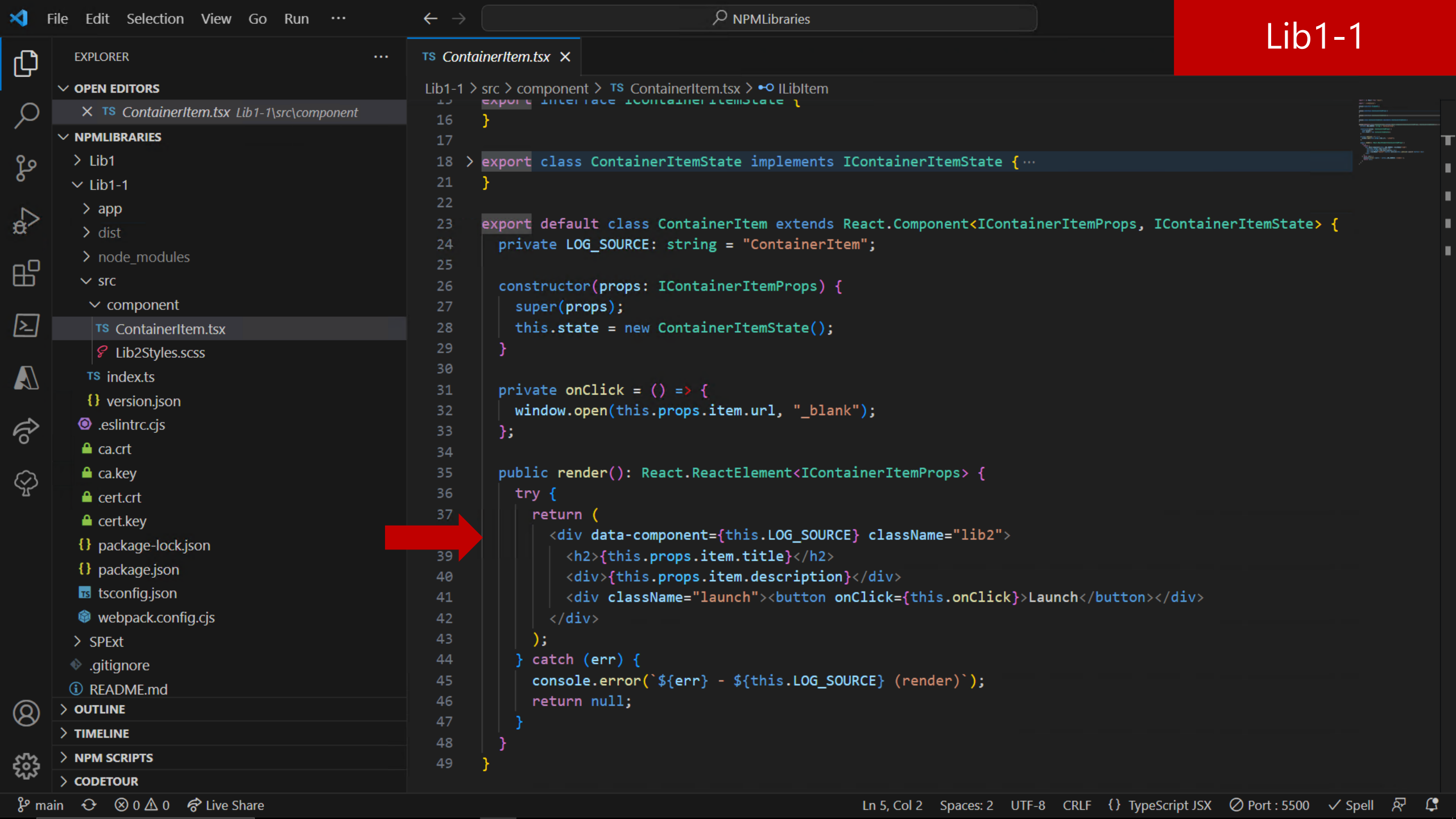
Lib1-1 > webpack.config.js > ...

```
4   module.exports = (env, argv) => {
17      devtool: 'inline-source-map',
18      resolve: {···
21      },
22      module: {···
45      },
46      externals: {···
49      },
50      devServer: {
51        compress: true,
52        hot: true,
53        port: 3000,
54        client: {
55          webSocketURL: 'wss://localhost:3000/ws'
56        },
57        https: {
            key: fs.readFileSync(path.resolve(__dirname, '../certs/cert.key')),
            cert: fs.readFileSync(path.resolve(__dirname, '../certs/cert.crt')),
60          ca: fs.readFileSync(path.resolve(__dirname, '../certs/ca.crt')),
61        },
62        static: [{
            directory: path.join(__dirname, 'app'),
            publicPath: "/"
65        },
66        {
67          directory: path.join(__dirname, "node_modules"),
68          publicPath: "/"
69        }
70        ],
71      }
72    };
73  };
74
```


File Edit Selection View Go Run ···
NPMLibraries
Lib1-1
EXPLORER
OPEN EDITORS
index.html Lib1-1\app
NPMLIBRARIES
Lib1
Lib1-1
app
index.html
dist
node_modules
src
.eslintrc.cjs
ca.crt
ca.key
cert.crt
cert.key
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index.html
Lib1-1 > app > index.html > ...
1 <!DOCTYPE HTML PUBLIC "-//W3C//DTD HTML 4.01 Transitional//EN" "http://www.w3.org/TR/html4/loose.dtd">
2 <html lang="en">
3
4 <head>
5 <meta http-equiv="content-type" content="text/html; charset=utf-8">
6 <title>Lib1_1 Library Workbench</title>
<script src="./react/umd/react.development.js"></script>
<script src="./react-dom/umd/react-dom.development.js"></script>
<script src="./lib1_1.js"></script>
10 </head>
11
12 <body>
13
14
15 </body>
16
17 <script>
18 const container = document.getElementById("lib1_1");
const element = React.createElement(
20 lib11.ContainerItem, {
21 key: 1,
22 item: {
23 title: "Microsoft's Home Site",
24 description: "The root Microsoft site.",
25 url: "https://microsoft.com"
26 }
27 }
28);
29
ReactDOM.render(element, container);
31 </script>
32
33 </html>
main 0 0 Live Share
Ln 1, Col 1 Spaces: 2 UTF-8 CRLF HTML Port : 5500 Spell

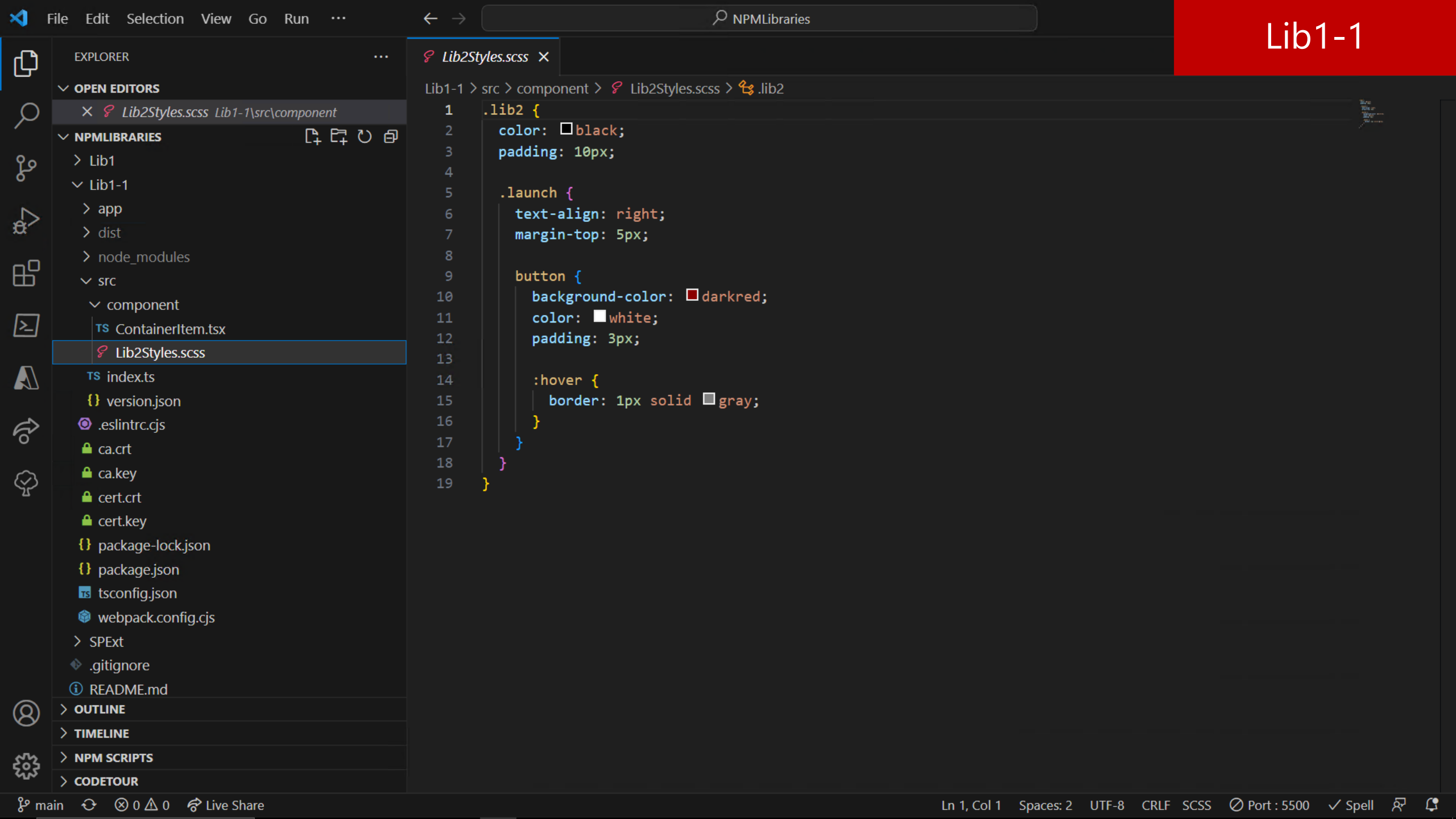

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index.ts
Lib1-1 > src > index.ts
1 import ContainerItem, { IContainerItemProps, ILibItem } from "./component/ContainerItem";
2
3 export type { IContainerItemProps, ILibItem };
4 export {ContainerItem};
5
main
0 0
Live Share
Ln 4, Col 23
Spaces: 2
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ContainerItem.tsx
Lib1-1 > src > component > ContainerItem.tsx > IContainerItemState
1 import * as React from "react";
2
3 import "./Lib2Styles";
4
5 export interface ILibItem {
6 title: string;
7 description: string;
8 url: string;
9 }
10
11 export interface IContainerItemProps {
12 item: ILibItem;
13 }
14
15 export interface IContainerItemState {
16 }
17
18 export class ContainerItemState implements IContainerItemState {
19 // eslint-disable-next-line @typescript-eslint/no-empty-function
20 constructor() { }
21 }
22
23 export default class ContainerItem extends React.Component<IContainerItemProps, IContainerItemState> {
49 }
main
0 0
Live Share
Ln 15, Col 39
Spaces: 2
UTF-8
CRLF
TypeScript JSX
Port : 5500
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ContainerItem.tsx
Lib1-1 > src > component > ContainerItem.tsx > ILibItem
}
export class ContainerItemState implements IContainerItemState {···
}
export default class ContainerItem extends React.Component<IContainerItemProps, IContainerItemState> {
private LOG_SOURCE: string = "ContainerItem";
constructor(props: IContainerItemProps) {
super(props);
this.state = new ContainerItemState();
}
private onClick = () => {
window.open(this.props.item.url, "_blank");
};
public render(): React.ReactElement<IContainerItemProps> {
try {
return (

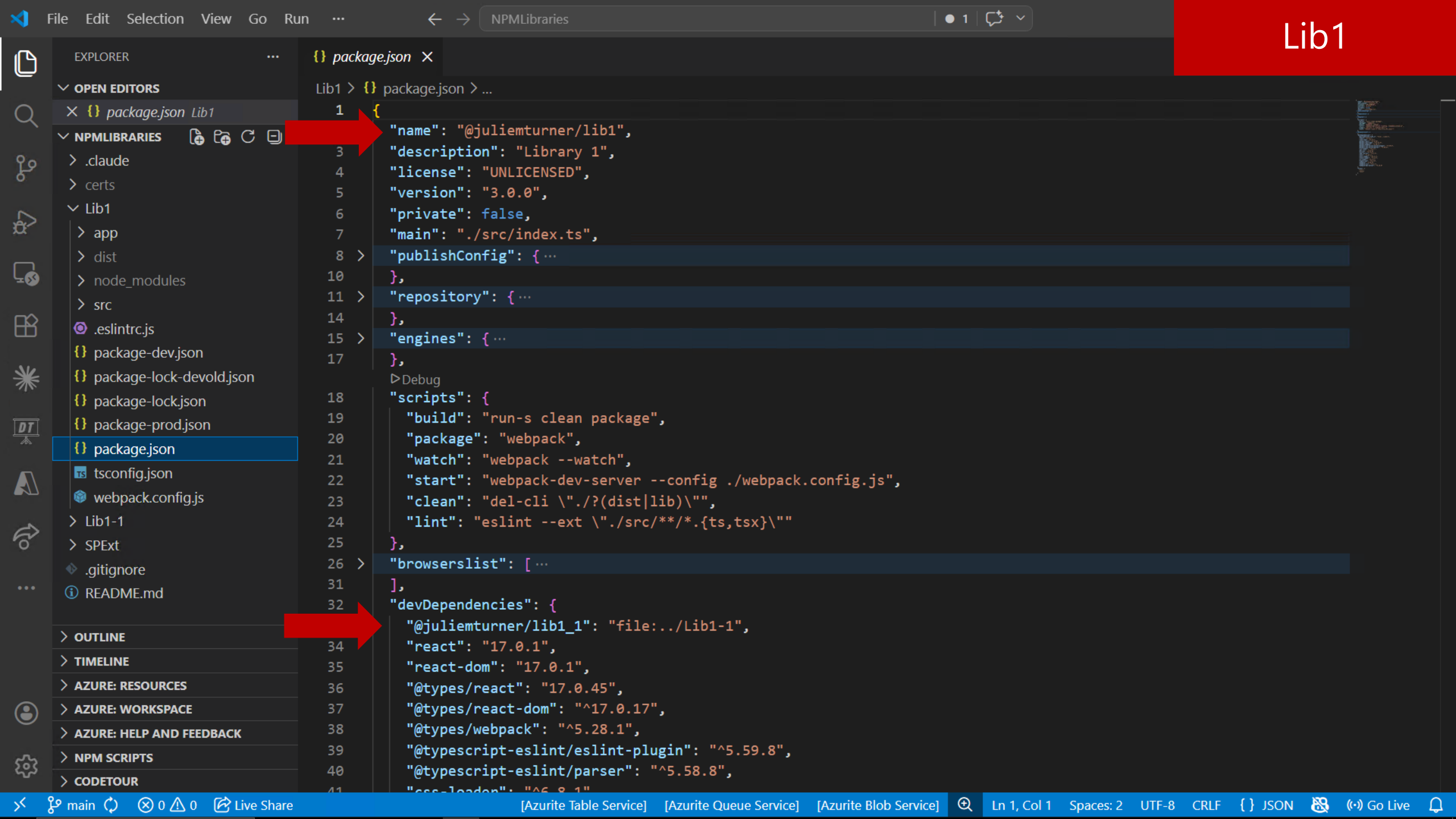
<h2>{this.props.item.title}</h2>
{this.props.item.description}
<button onClick={this.onClick}>Launch</button>

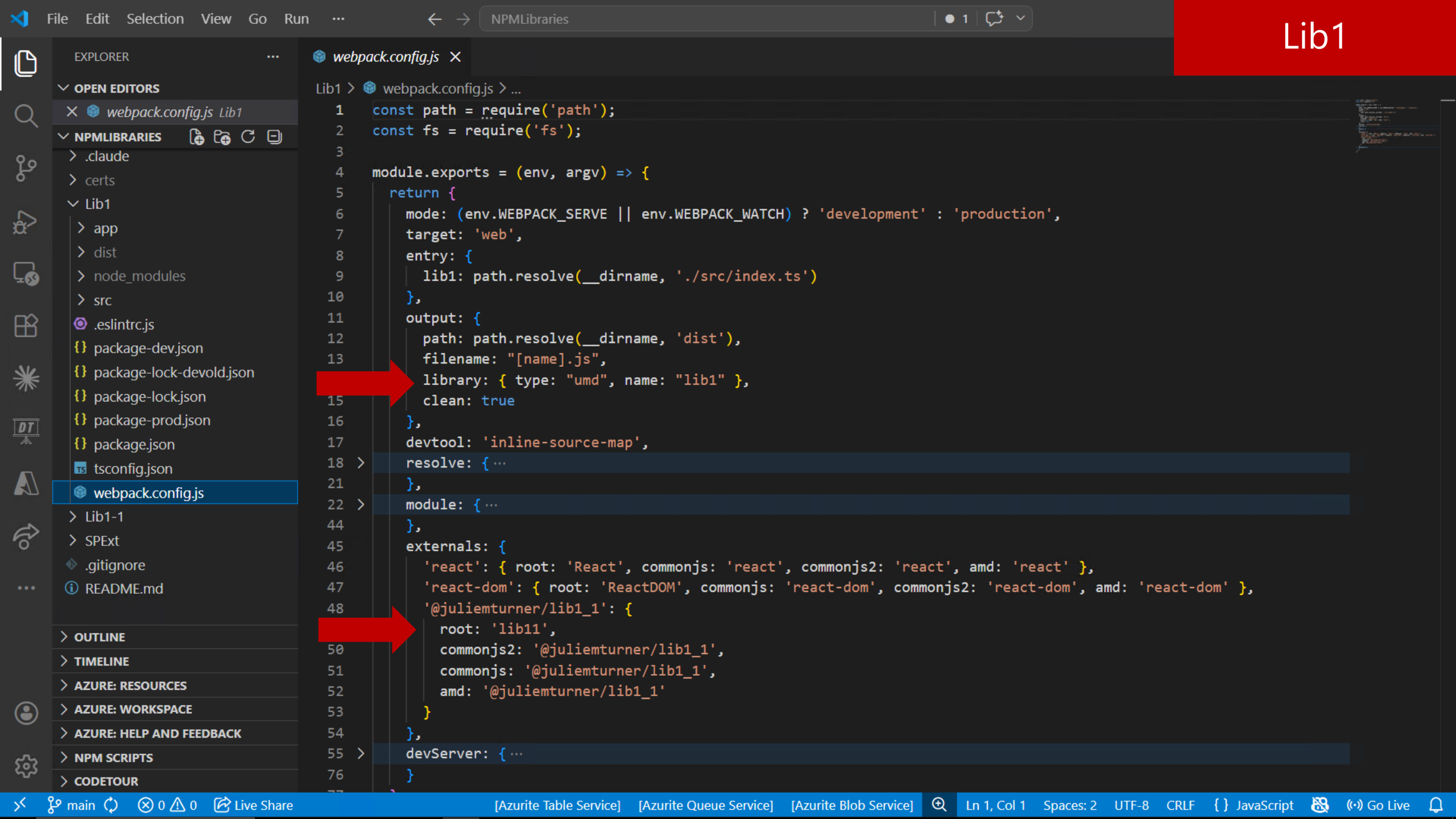
);
} catch (err) {
console.error(`${err} - ${this.LOG_SOURCE} (render)`);
return null;
}
}
}
main
0 0
Live Share
Ln 5, Col 2
Spaces: 2
UTF-8
CRLF
TypeScript JSX
Port : 5500
Spell

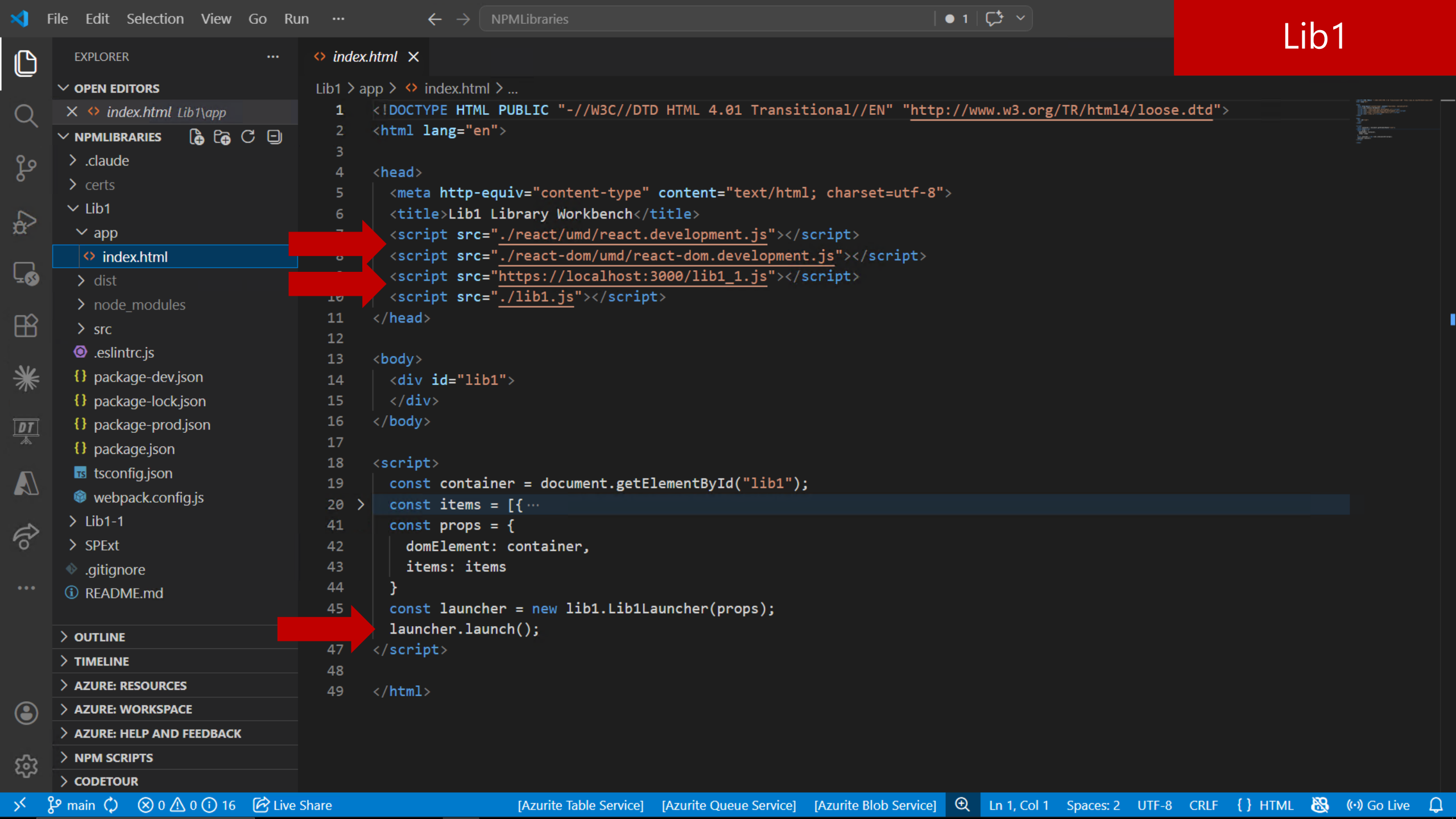

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Lib2Styles.scss
Lib1-1 > src > component > Lib2Styles.scss > .lib2
.lib2 {
color: black;
padding: 10px;
.launch {
text-align: right;
margin-top: 5px;
button {
background-color: darkred;
color: white;
padding: 3px;
:hover {
border: 1px solid gray;
}
}
}
}
main
0 0
Live Share
Ln 1, Col 1
Spaces: 2
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SCSS
Port : 5500
Spell

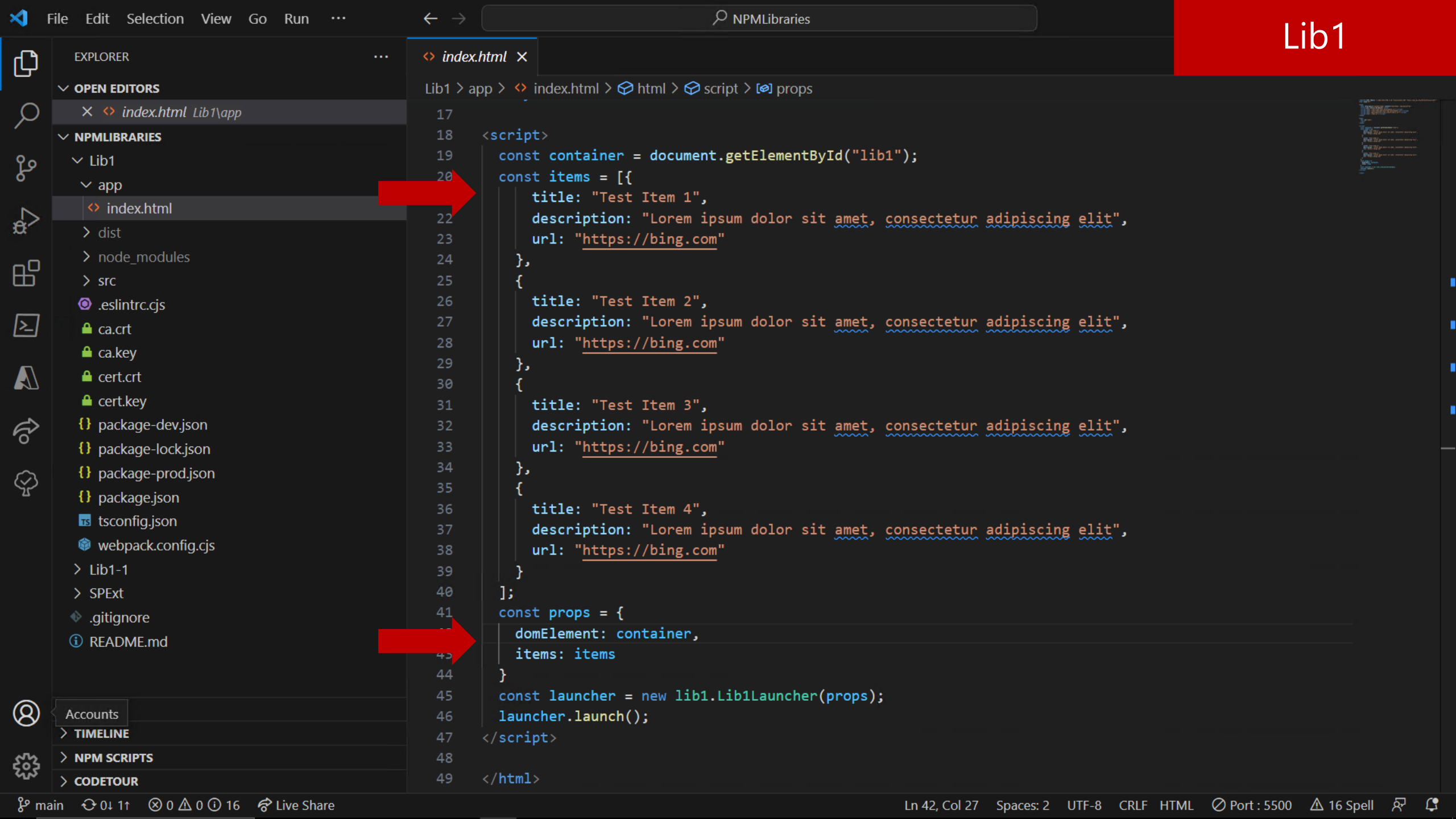
Menu Item Container

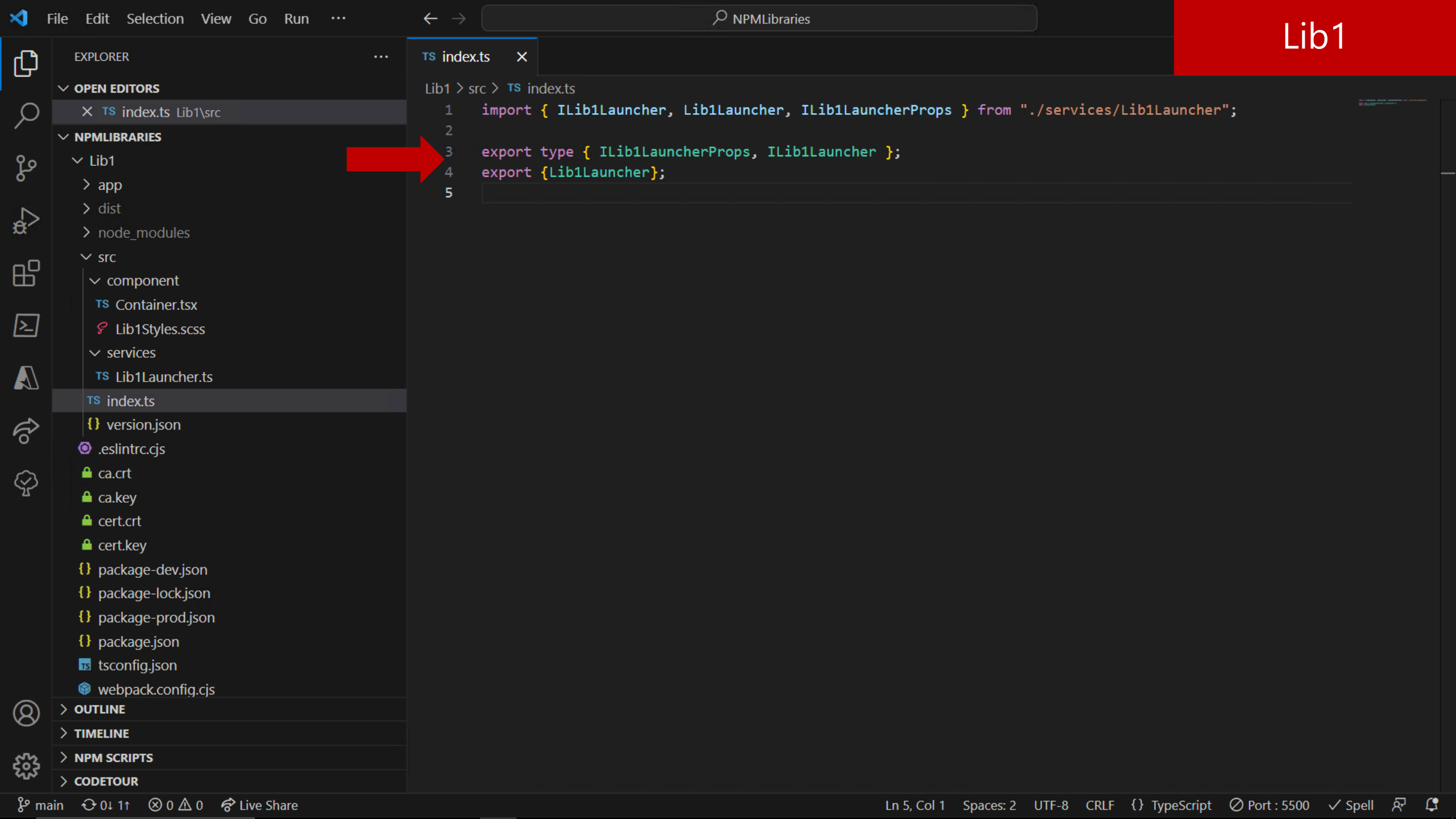
Lib1

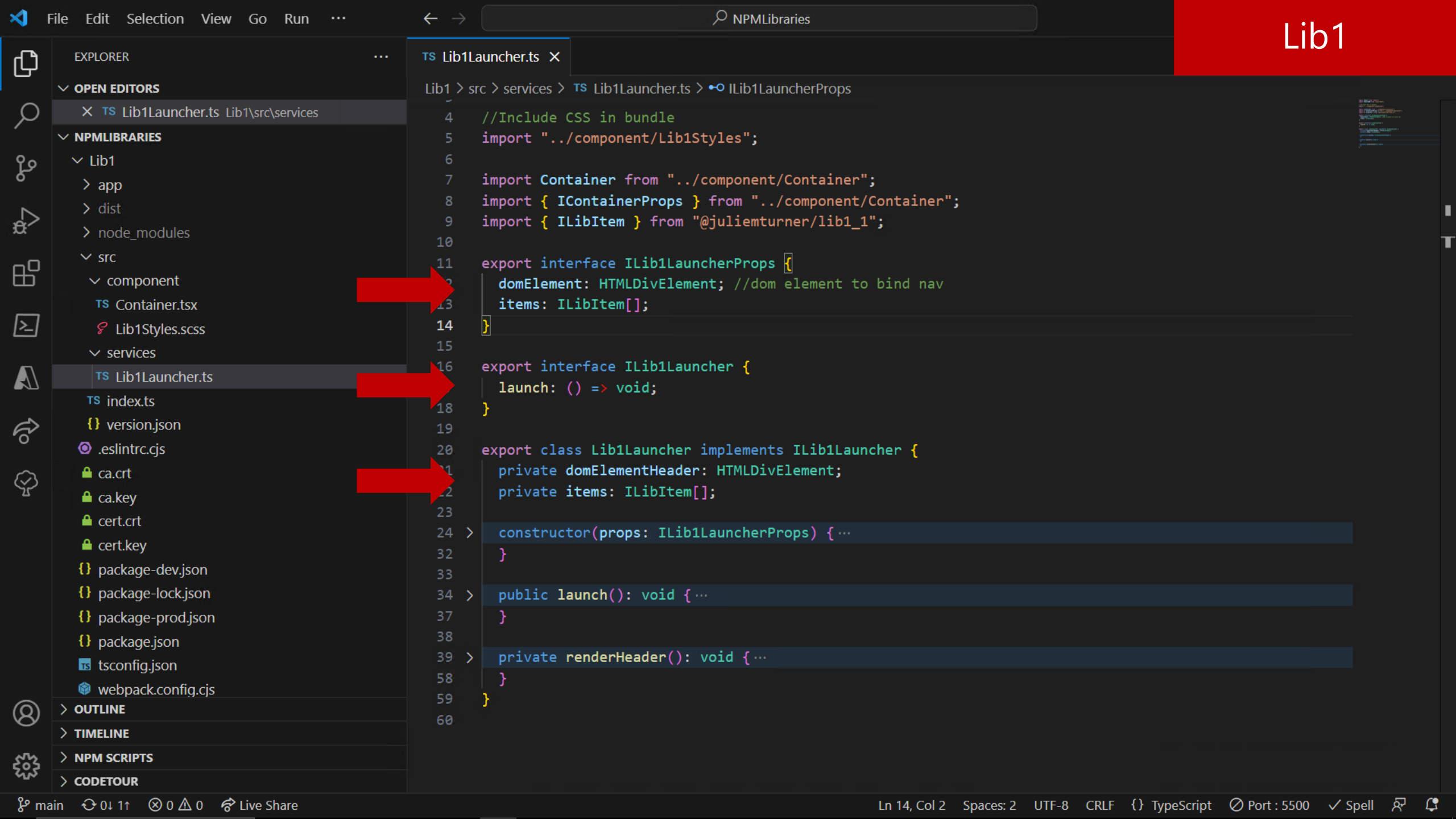

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package.json
Lib1 > package.json > ...
{
"name": "@juliemturner/lib1",
"description": "Library 1",
"license": "UNLICENSED",
"version": "3.0.0",
"private": false,
"main": "./src/index.ts",
"publishConfig": { ···
},
"repository": { ···
},
"engines": { ···
},
Debug
"scripts": {
"build": "run-s clean package",
"package": "webpack",
"watch": "webpack --watch",
"start": "webpack-dev-server --config ./webpack.config.js",
"clean": "del-cli \"./?(dist|lib)\"",
"lint": "eslint --ext \"./src/**/*.{ts,tsx}\""
},
"browserslist": [···
],
"devDependencies": {
"@juliemturner/lib1_1": "file:../Lib1-1",
"react": "17.0.1",
"react-dom": "17.0.1",
"@types/react": "17.0.45",
"@types/react-dom": "^17.0.17",
"@types/webpack": "^5.28.1",
"@typescript-eslint/eslint-plugin": "^5.59.8",
"@typescript-eslint/parser": "^5.58.8",
main
0 0
Live Share
[Azurite Table Service]
[Azurite Queue Service]
[Azurite Blob Service]
Ln 1, Col 1
Spaces: 2
UTF-8
CRLF
{ } JSON
Go Live
Lib1

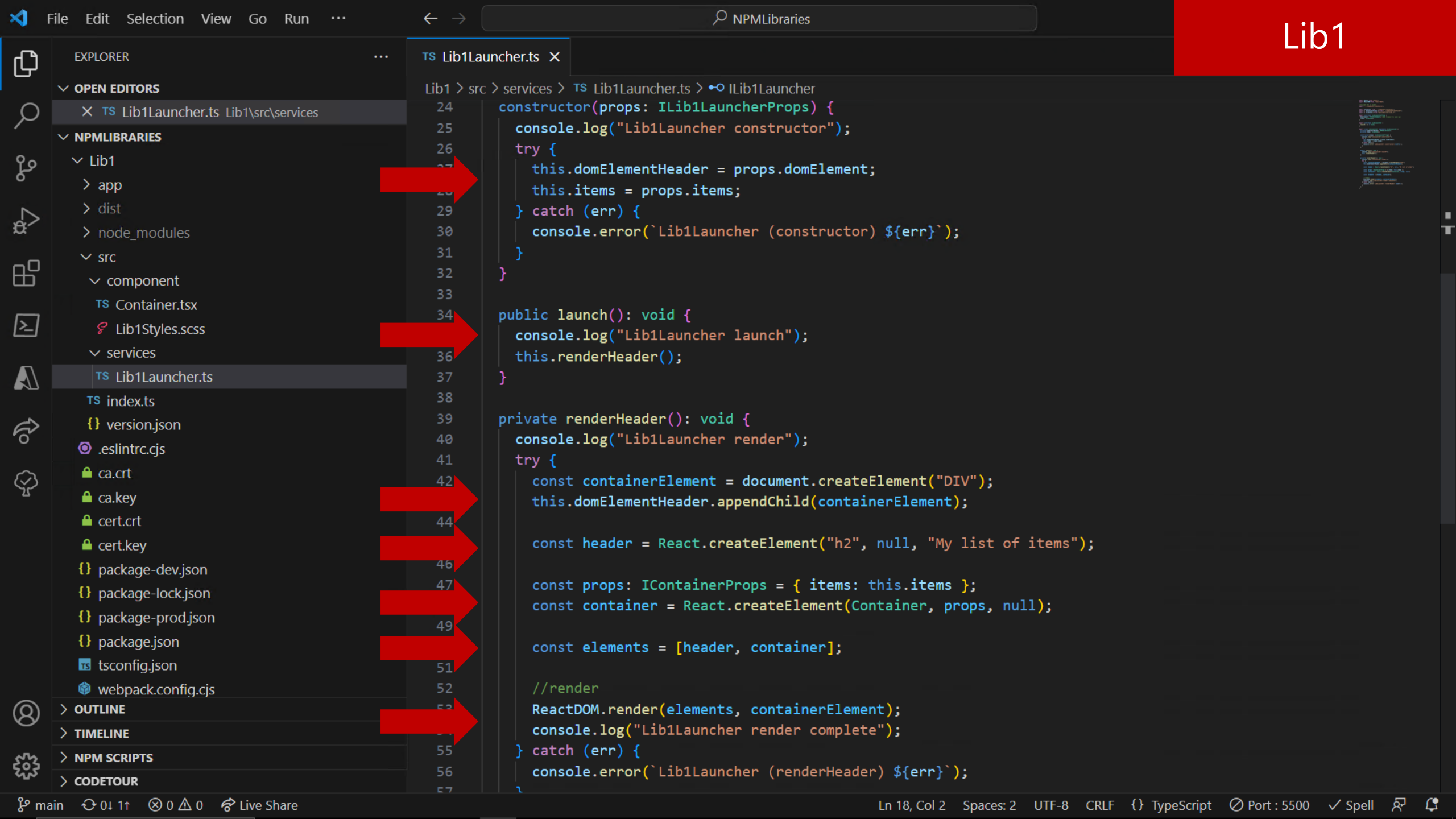

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Lib1 > webpack.config.js > ...
1 const path = require('path');
2 const fs = require('fs');
3
4 module.exports = (env, argv) => {
5 return {
6 mode: (env.WEBPACK_SERVE || env.WEBPACK_WATCH) ? 'development' : 'production',
7 target: 'web',
8 entry: {
9 lib1: path.resolve(__dirname, './src/index.ts')
10 },
11 output: {
12 path: path.resolve(__dirname, 'dist'),
13 filename: "[name].js",
library: { type: "umd", name: "lib1" },
15 clean: true
16 },
17 devtool: 'inline-source-map',
18 resolve: {···
21 },
22 module: {···
44 },
45 externals: {
46 'react': { root: 'React', commonjs: 'react', commonjs2: 'react', amd: 'react' },
47 'react-dom': { root: 'ReactDOM', commonjs: 'react-dom', commonjs2: 'react-dom', amd: 'react-dom' },
48 '@juliemturner/lib1_1': {
root: 'lib11',
50 commonjs2: '@juliemturner/lib1_1',
51 commonjs: '@juliemturner/lib1_1',
52 amd: '@juliemturner/lib1_1'
53 }
54 },
55 devServer: {···
76 }
main 0 0 Live Share [Azurite Table Service] [Azurite Queue Service] [Azurite Blob Service] Ln 1, Col 1 Spaces: 2 UTF-8 CRLF { } JavaScript Go Live

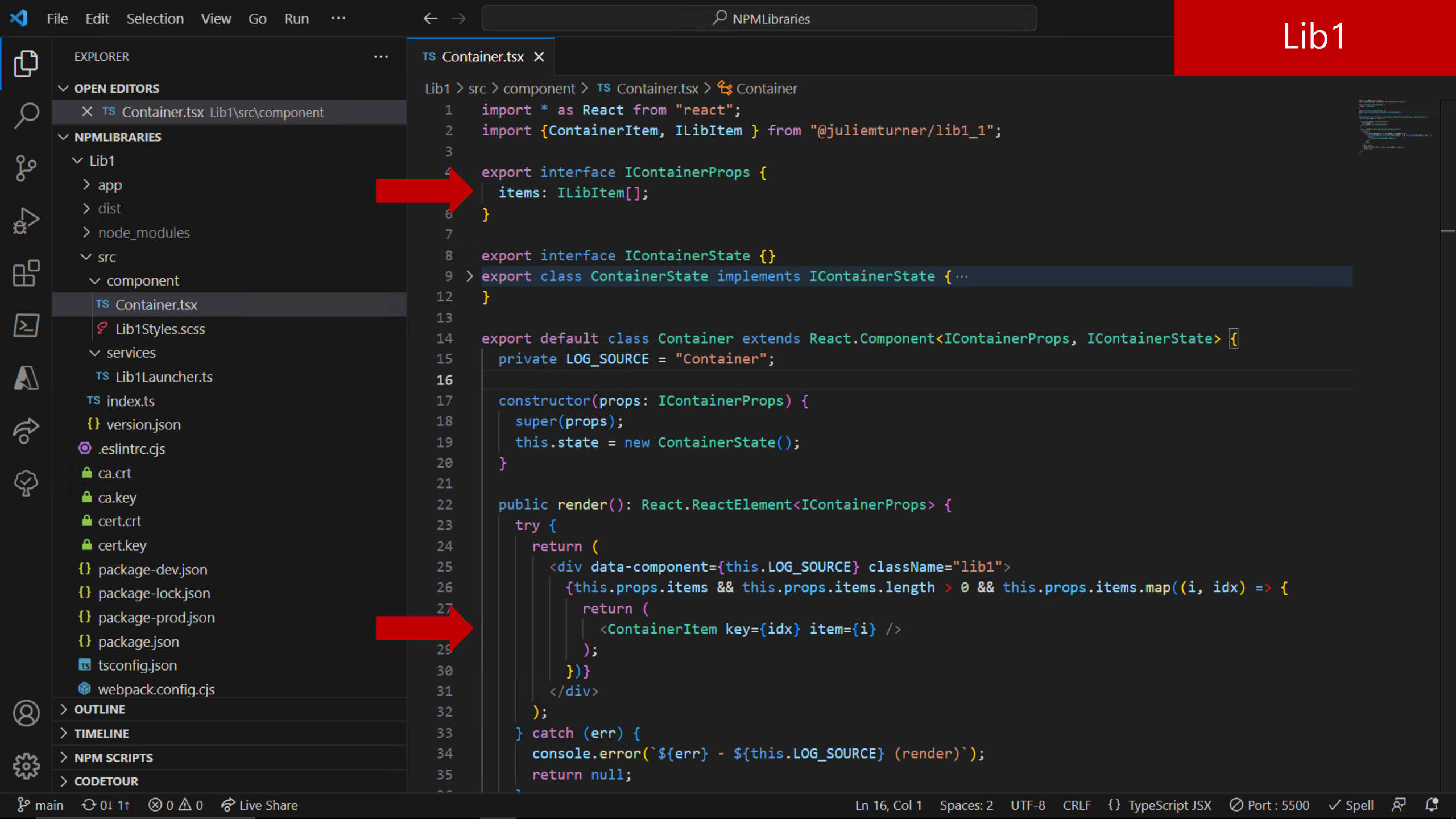

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Lib1 > app > index.html > ...
1 <!DOCTYPE HTML PUBLIC "-//W3C//DTD HTML 4.01 Transitional//EN" "http://www.w3.org/TR/html4/loose.dtd">
2 <html lang="en">
3
4 <head>
5 <meta http-equiv="content-type" content="text/html; charset=utf-8">
6 <title>Lib1 Library Workbench</title>
7 <script src="./react/umd/react.development.js"></script>
8 <script src="./react-dom/umd/react-dom.development.js"></script>
9 <script src="https://localhost:3000/lib1_1.js"></script>
10 <script src="./lib1.js"></script>
11 </head>
12
13 <body>
14
15
16 </body>
17
18 <script>
19 const container = document.getElementById("lib1");
20 const items = [{···
41 const props = {
42 domElement: container,
43 items: items
44 }
45 const launcher = new lib1.Lib1Launcher(props);
46 launcher.launch();
47 </script>
48
49 </html>
main
0 0 16
Live Share
[Azurite Table Service]
[Azurite Queue Service]
[Azurite Blob Service]
Ln 1, Col 1
Spaces: 2
UTF-8
CRLF
{ } HTML
Go Live

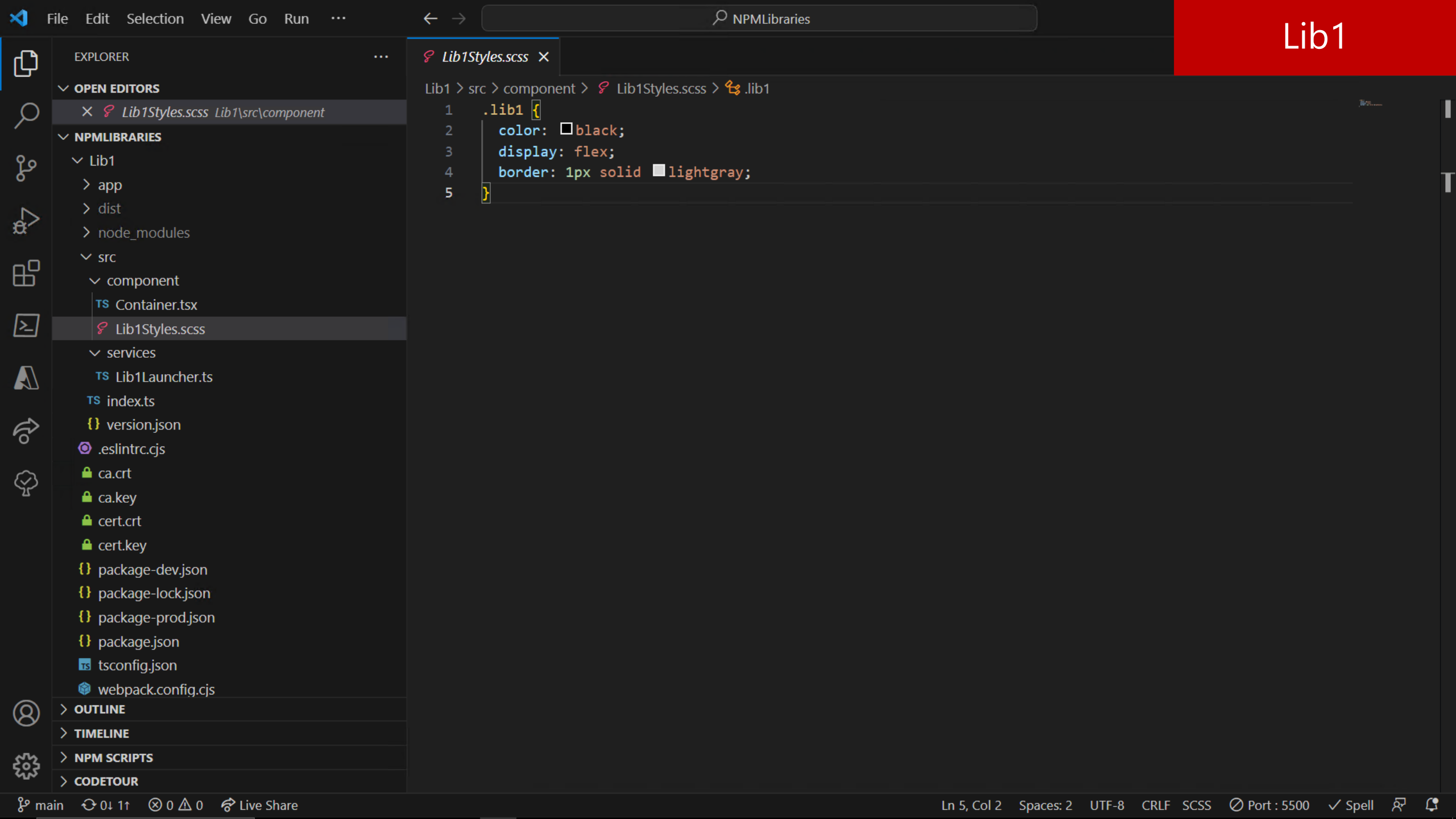

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index.html
Lib1 > app > index.html > html > script > props
<script>
const container = document.getElementById("lib1");
const items = [{
title: "Test Item 1",
description: "Lorem ipsum dolor sit amet, consectetur adipiscing elit",
url: "https://bing.com"
},
{
title: "Test Item 2",
description: "Lorem ipsum dolor sit amet, consectetur adipiscing elit",
url: "https://bing.com"
},
{
title: "Test Item 3",
description: "Lorem ipsum dolor sit amet, consectetur adipiscing elit",
url: "https://bing.com"
},
{
title: "Test Item 4",
description: "Lorem ipsum dolor sit amet, consectetur adipiscing elit",
url: "https://bing.com"
}
];
const props = {
domElement: container,
items: items
}
const launcher = new lib1.Lib1Launcher(props);
launcher.launch();
</script>
</html>
main 0↓ 1↑ 0 0 16 Live Share
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index.ts
Lib1 > src > index.ts
1 import { ILib1Launcher, Lib1Launcher, ILib1LauncherProps } from "./services/Lib1Launcher";
2
3 export type { ILib1LauncherProps, ILib1Launcher };
4 export {Lib1Launcher};
5
main 0↓ 1↑ 0 0 Live Share
Ln 5, Col 1 Spaces: 2 UTF-8 CRLF {} TypeScript Port : 5500 Spell


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Lib1Launcher.ts
Lib1 > src > services > Lib1Launcher.ts > ILib1LauncherProps
4 //Include CSS in bundle
5 import "../component/Lib1Styles";
6
7 import Container from "../component/Container";
8 import { IContainerProps } from "../component/Container";
9 import { ILibItem } from "@juliemturner/lib1_1";
10
11 export interface ILib1LauncherProps {
13 domElement: HTMLDivElement; //dom element to bind nav
14 items: ILibItem[];
15
16 export interface ILib1Launcher {
18 launch: () => void;
19
20 export class Lib1Launcher implements ILib1Launcher {
21 private domElementHeader: HTMLDivElement;
22 private items: ILibItem[];
23
24 constructor(props: ILib1LauncherProps) {···
32 }
33
34 public launch(): void {···
37 }
38
39 private renderHeader(): void {···
58 }
59 }
60
main 0↓ 1↑ 0 0 Live Share
Ln 14, Col 2 Spaces: 2 UTF-8 CRLF TypeScript Port : 5500 Spell

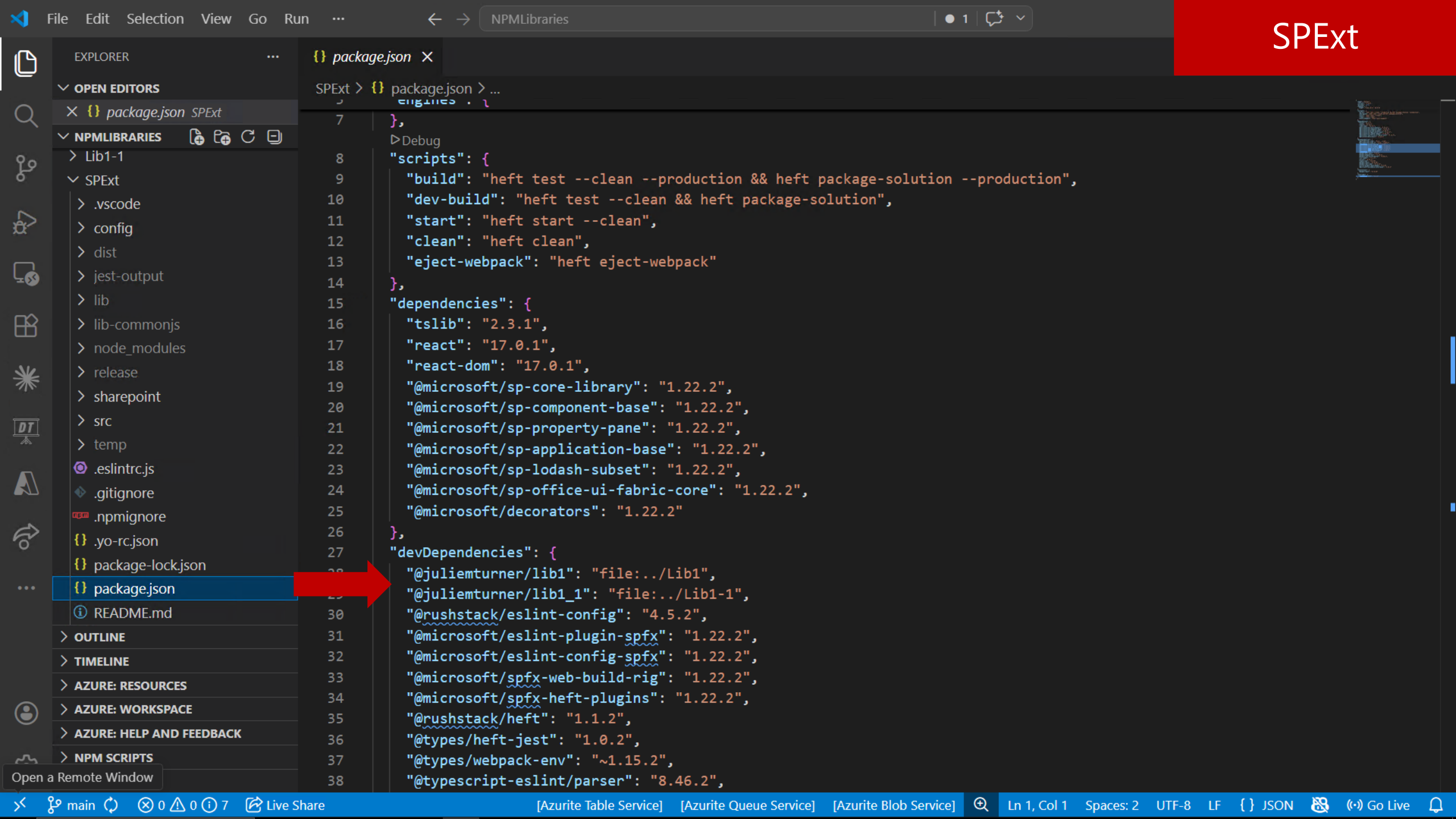

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Lib1Launcher.ts
Lib1 > src > services > Lib1Launcher.ts > ILib1Launcher
constructor(props: ILib1LauncherProps) {
console.log("Lib1Launcher constructor");
try {
this.domElementHeader = props.domElement;
this.items = props.items;
} catch (err) {
console.error(`Lib1Launcher (constructor) ${err}`);
}
}
public launch(): void {
console.log("Lib1Launcher launch");
this.renderHeader();
}
private renderHeader(): void {
console.log("Lib1Launcher render");
try {
const containerElement = document.createElement("DIV");
this.domElementHeader.appendChild(containerElement);
const header = React.createElement("h2", null, "My list of items");
const props: IContainerProps = { items: this.items };
const container = React.createElement(Container, props, null);
const elements = [header, container];
//render
ReactDOM.render(elements, containerElement);
console.log("Lib1Launcher render complete");
} catch (err) {
console.error(`Lib1Launcher (renderHeader) ${err}`);
main
0↓ 1↑
0 0
Live Share
Ln 18, Col 2
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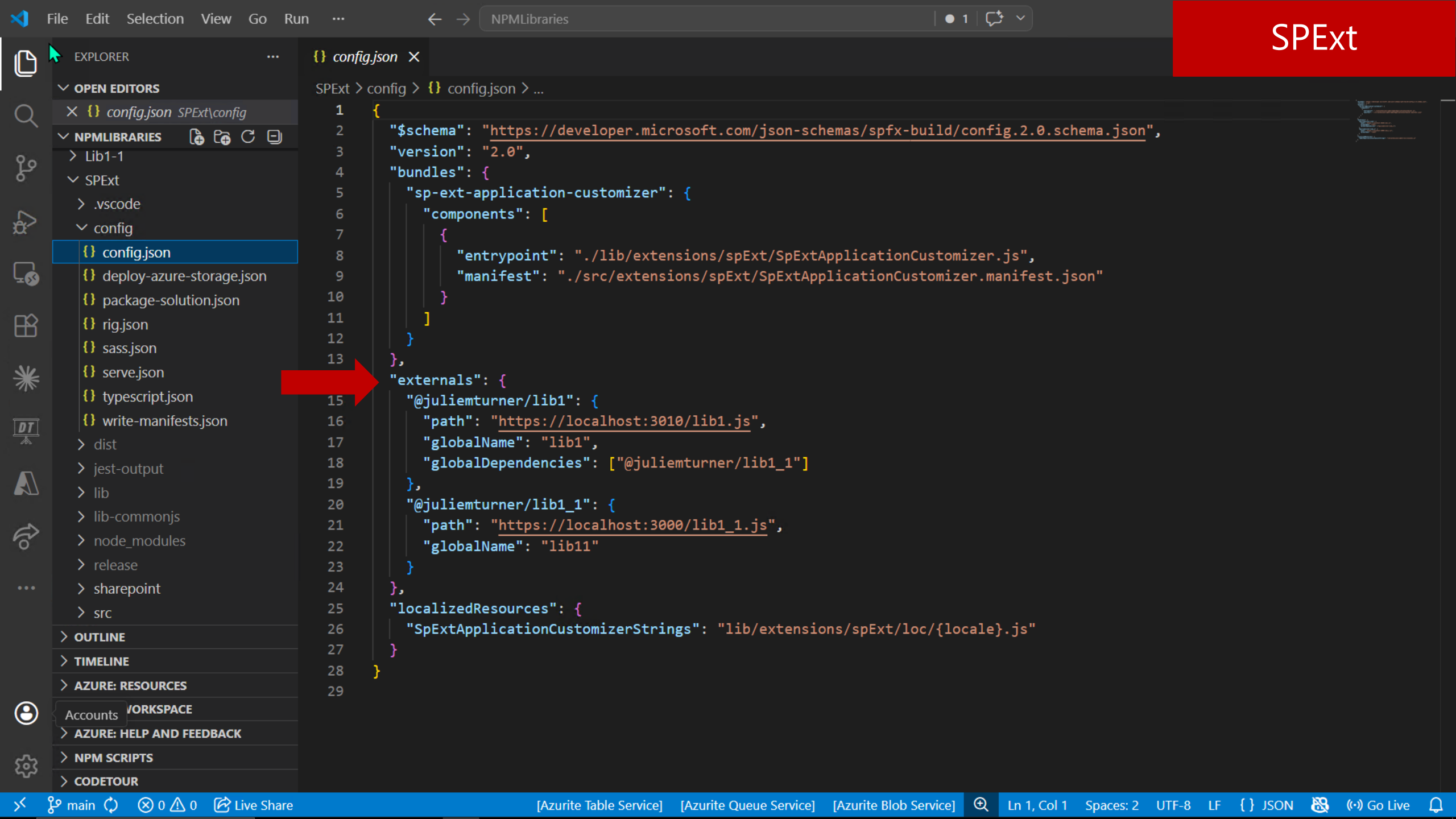

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package.json
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webpack.config.cjs
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Container.tsx
Lib1 > src > component > Container.tsx > Container
1 import * as React from "react";
2 import {ContainerItem, ILibItem } from "@juliemturner/lib1_1";
3
4 export interface IContainerProps {
5 items: ILibItem[];
6 }
7
8 export interface IContainerState {}
9 export class ContainerState implements IContainerState {···
12 }
13
14 export default class Container extends React.Component<IContainerProps, IContainerState> {
15 private LOG_SOURCE = "Container";
16
17 constructor(props: IContainerProps) {
18 super(props);
19 this.state = new ContainerState();
20 }
21
22 public render(): React.ReactElement<IContainerProps> {
23 try {
24 return (
25
26 {this.props.items && this.props.items.length > 0 && this.props.items.map((i, idx) => {
27 return (
<ContainerItem key={idx} item={i} />
);
})}
31
32);
33 } catch (err) {
34 console.error(`${err} - ${this.LOG_SOURCE} (render)`);
35 return null;
main 0↓ 1↑ 0 0 Live Share
Ln 16, Col 1 Spaces: 2 UTF-8 CRLF {} TypeScript JSX Port : 5500 Spell

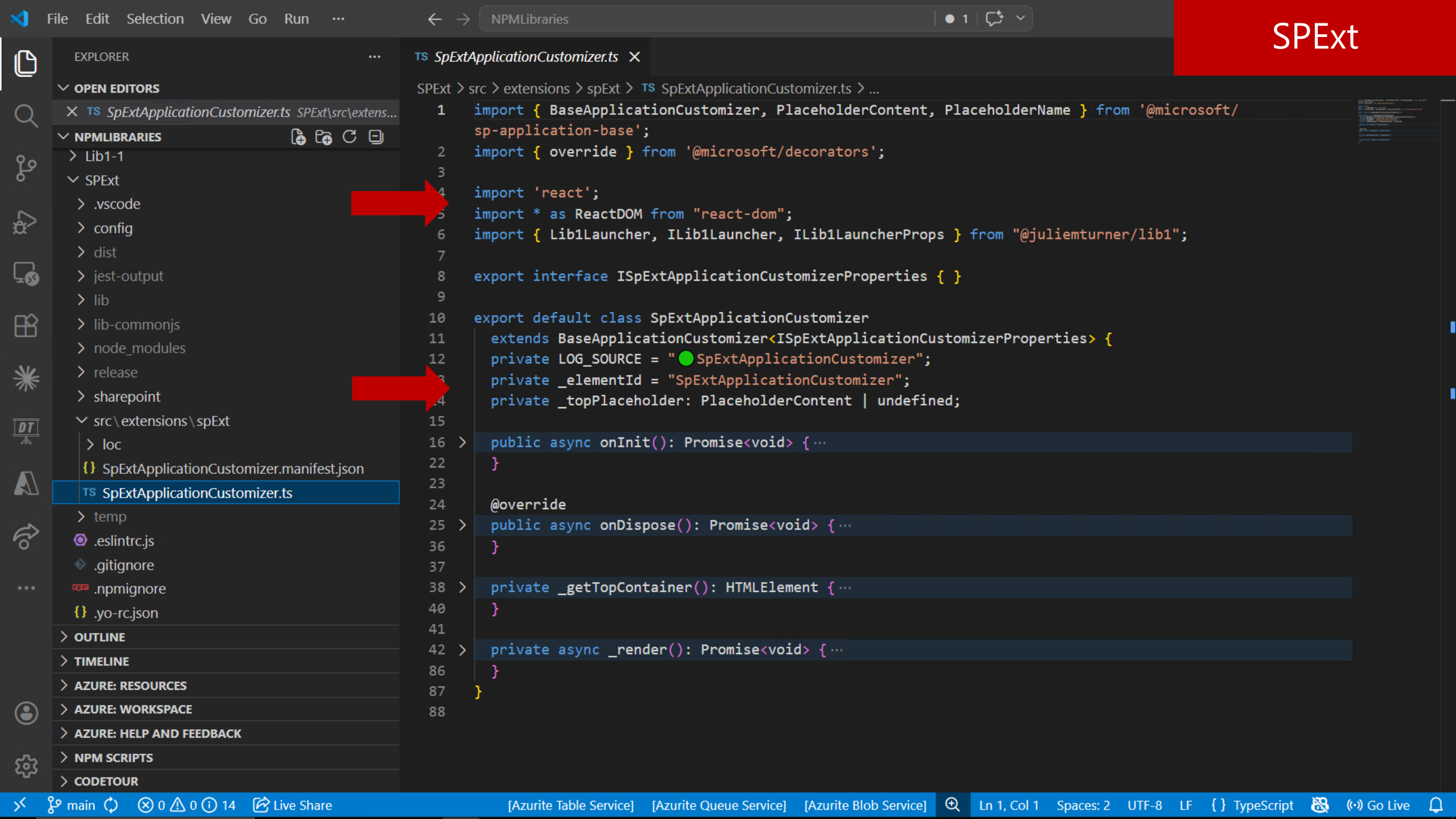

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NPMLibraries
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Lib1Styles.scss Lib1\src\component
NPMLIBRARIES
Lib1
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Container.tsx
Lib1Styles.scss
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Lib1Launcher.ts
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ca.crt
ca.key
cert.crt
cert.key
package-dev.json
package-lock.json
package-prod.json
package.json
tsconfig.json
webpack.config.cjs
OUTLINE
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CODETOUR
Lib1Styles.scss
Lib1 > src > component > Lib1Styles.scss > .lib1
1 .lib1 {
2 color: black;
3 display: flex;
4 border: 1px solid lightgray;
5 }
main 0↓ 1↑ 0 0 Live Share
Ln 5, Col 2 Spaces: 2 UTF-8 CRLF SCSS Port : 5500 Spell

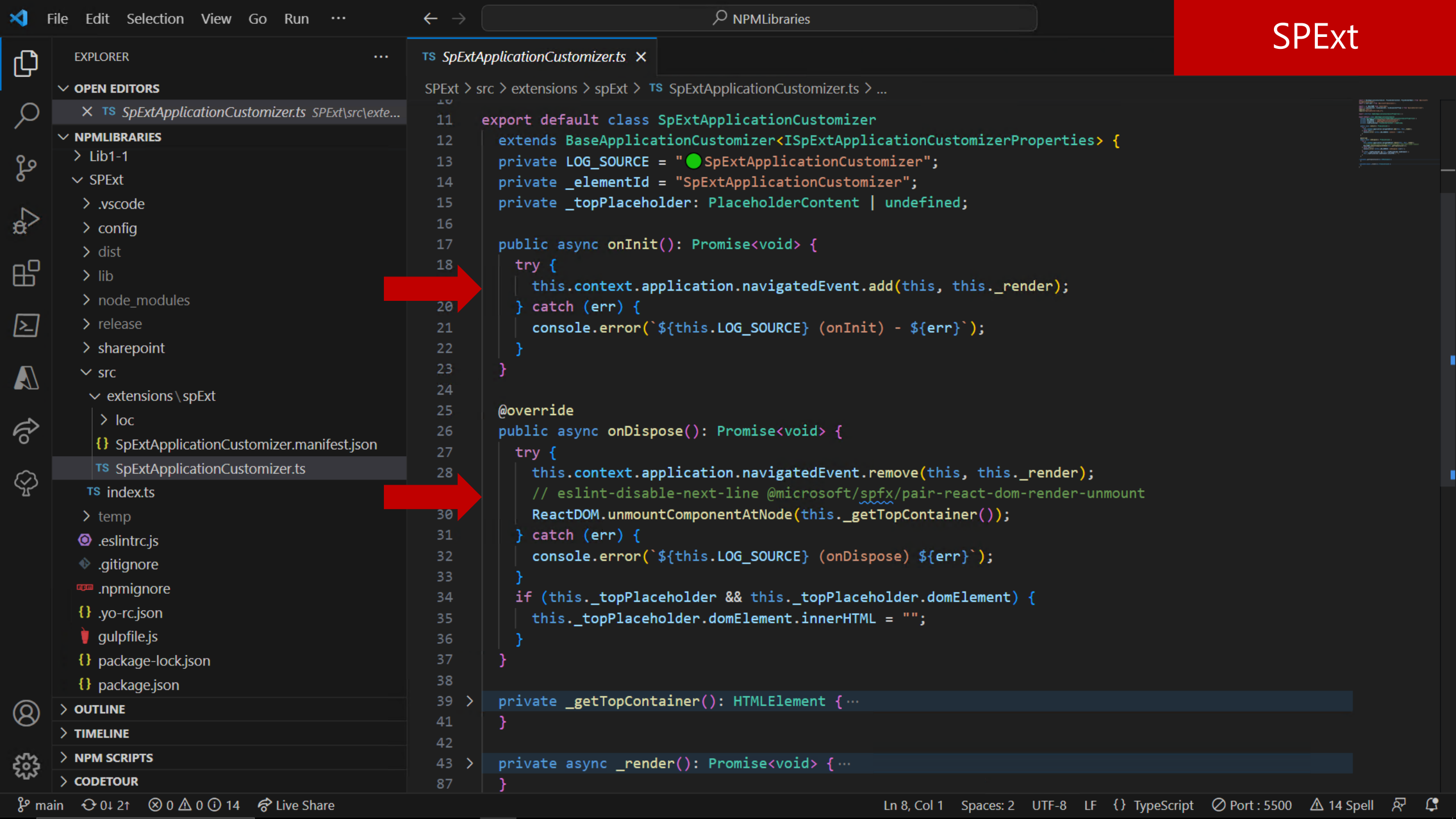
Bootstrap Menu Into SharePoint

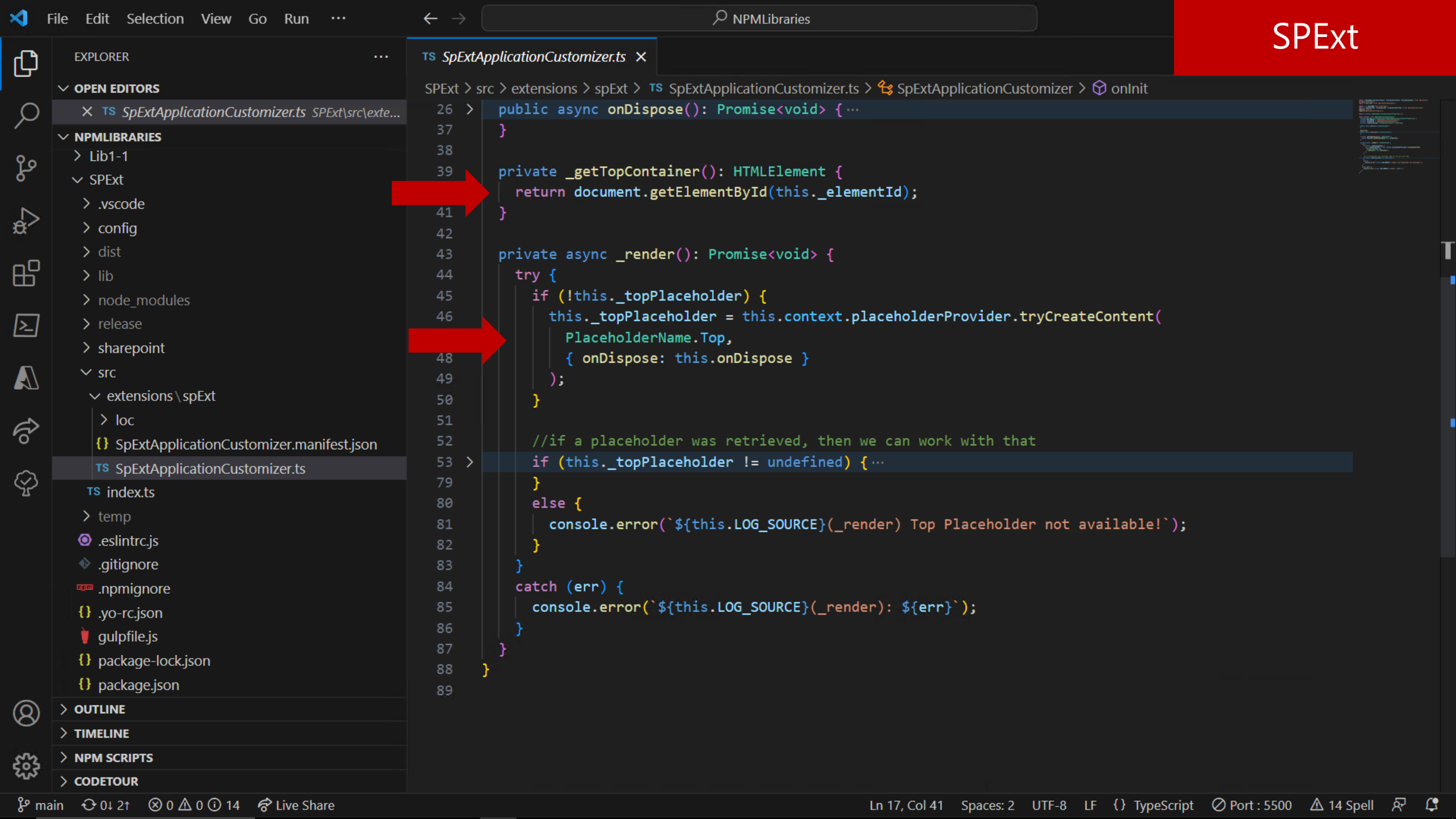
SPFx

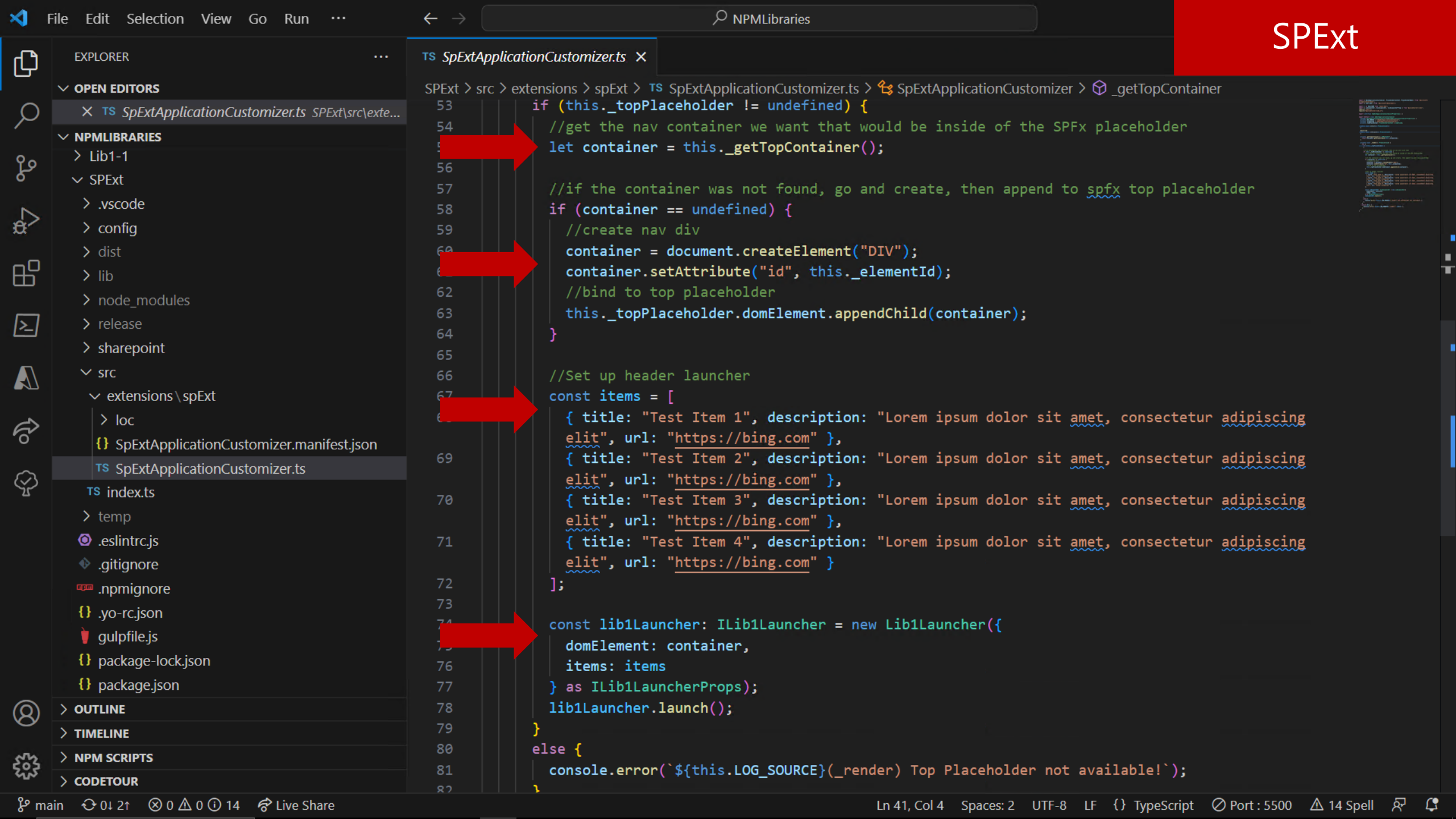

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NPMLibraries
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AZURE: HELP AND FEEDBACK
NPM SCRIPTS
Open a Remote Window
package.json
SPExt > package.json > ...
},
Debug
"scripts": {
"build": "heft test --clean --production && heft package-solution --production",
"dev-build": "heft test --clean && heft package-solution",
"start": "heft start --clean",
"clean": "heft clean",
"eject-webpack": "heft eject-webpack"
},
"dependencies": {
"tslib": "2.3.1",
"react": "17.0.1",
"react-dom": "17.0.1",
"@microsoft/sp-core-library": "1.22.2",
"@microsoft/sp-component-base": "1.22.2",
"@microsoft/sp-property-pane": "1.22.2",
"@microsoft/sp-application-base": "1.22.2",
"@microsoft/sp-lodash-subset": "1.22.2",
"@microsoft/sp-office-ui-fabric-core": "1.22.2",
"@microsoft/decorators": "1.22.2"
},
"devDependencies": {
"@juliemturner/lib1": "file:../Lib1",
"@juliemturner/lib1_1": "file:../Lib1-1",
"@rushstack/eslint-config": "4.5.2",
"@microsoft/eslint-plugin-spfx": "1.22.2",
"@microsoft/eslint-config-spfx": "1.22.2",
"@microsoft/spfx-web-build-rig": "1.22.2",
"@microsoft/spfx-heft-plugins": "1.22.2",
"@rushstack/heft": "1.1.2",
"@types/heft-jest": "1.0.2",
"@types/webpack-env": "~1.15.2",
"@typescript-eslint/parser": "8.46.2",
main 0 0 7 Live Share
[Azurite Table Service] [Azurite Queue Service] [Azurite Blob Service]
Ln 1, Col 1 Spaces: 2 UTF-8 LF { } JSON Go Live

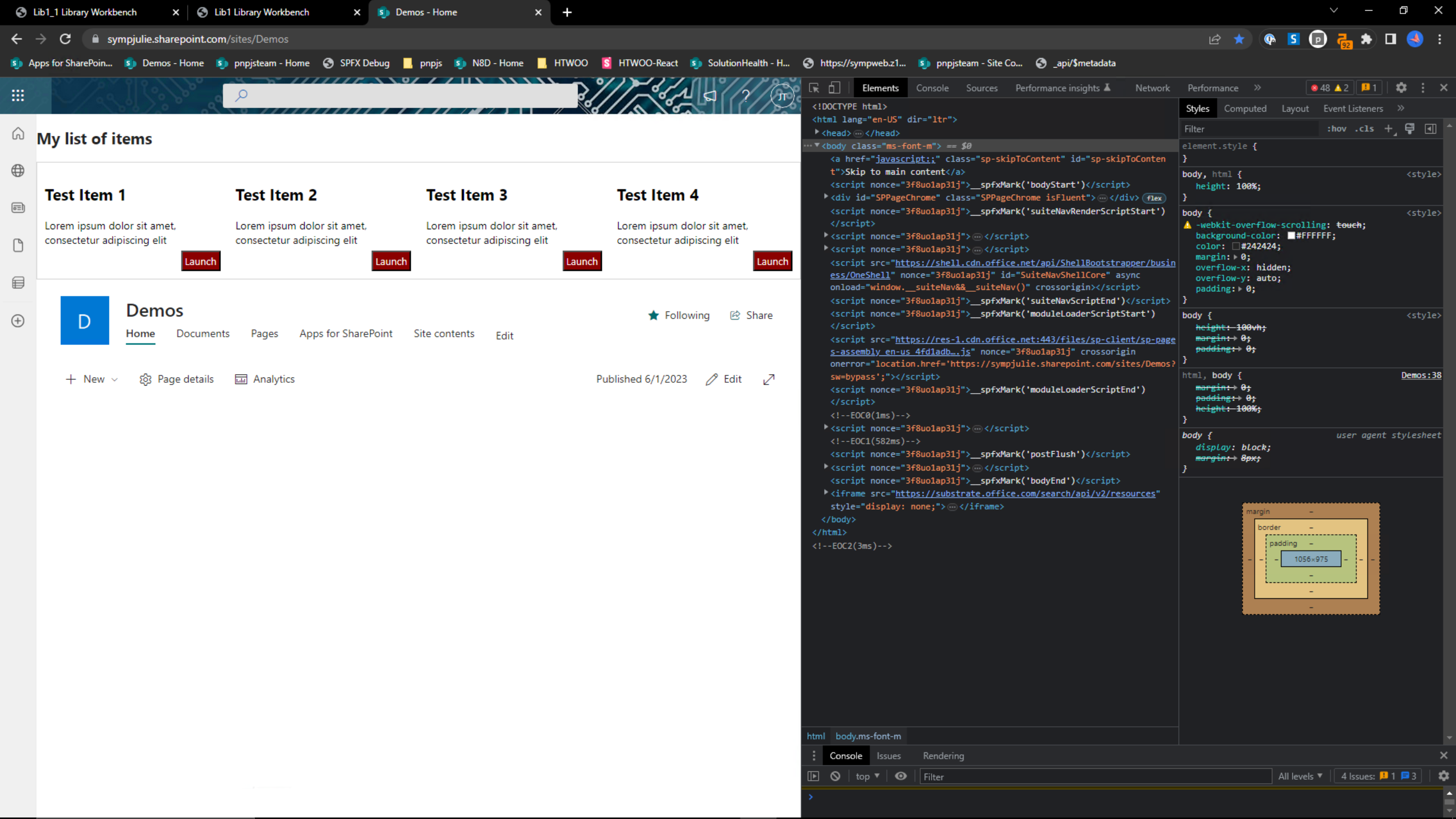

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AZURE: HELP AND FEEDBACK
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config.json
SPExt > config > config.json > ...
{
"$schema": "https://developer.microsoft.com/json-schemas/spfx-build/config.2.0.schema.json",
"version": "2.0",
"bundles": {
"sp-ext-application-customizer": {
"components": [
{
"entrypoint": "./lib/extensions/spExt/SpExtApplicationCustomizer.js",
"manifest": "./src/extensions/spExt/SpExtApplicationCustomizer.manifest.json"
}
]
}
},
"externals": {
"@juliemturner/lib1": {
"path": "https://localhost:3010/lib1.js",
"globalName": "lib1",
"globalDependencies": ["@juliemturner/lib1_1"]
},
"@juliemturner/lib1_1": {
"path": "https://localhost:3000/lib1_1.js",
"globalName": "lib11"
}
},
"localizedResources": {
"SpExtApplicationCustomizerStrings": "lib/extensions/spExt/loc/{locale}.js"
}
}
main
0 0
Live Share
[Azurite Table Service]
[Azurite Queue Service]
[Azurite Blob Service]
Ln 1, Col 1
Spaces: 2
UTF-8
LF
{ } JSON
Go Live


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.npmignore
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CODETOUR
SpExtApplicationCustomizer.ts
SPExt > src > extensions > spExt > SpExtApplicationCustomizer.ts > ...
1 import { BaseApplicationCustomizer, PlaceholderContent, PlaceholderName } from '@microsoft/sp-application-base';
2 import { override } from '@microsoft/decorators';
3
4 import 'react';
5 import * as ReactDOM from "react-dom";
6 import { Lib1Launcher, ILib1Launcher, ILib1LauncherProps } from "@juliemturner/lib1";
7
8 export interface ISpExtApplicationCustomizerProperties { }
9
10 export default class SpExtApplicationCustomizer
11 extends BaseApplicationCustomizer<ISpExtApplicationCustomizerProperties> {
12 private LOG_SOURCE = "SpExtApplicationCustomizer";
13 private _elementId = "SpExtApplicationCustomizer";
14 private _topPlaceholder: PlaceholderContent | undefined;
15
16 public async onInit(): Promise<void> {···
22 }
23
24 @override
25 public async onDispose(): Promise<void> {···
36 }
37
38 private _getTopContainer(): HTMLElement {···
40 }
41
42 private async _render(): Promise<void> {···
86 }
87 }
88
main 0 0 14 Live Share [Azurite Table Service] [Azurite Queue Service] [Azurite Blob Service] Ln 1, Col 1 Spaces: 2 UTF-8 LF TypeScript Go Live


File Edit Selection View Go Run ···
NPMLibraries
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SpExtApplicationCustomizer.ts
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.gitignore
.npmignore
.yo-rc.json
gulpfile.js
package-lock.json
package.json
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NPM SCRIPTS
CODETOUR
SpExtApplicationCustomizer.ts
SPExt > src > extensions > spExt > SpExtApplicationCustomizer.ts > ...
export default class SpExtApplicationCustomizer
extends BaseApplicationCustomizer<ISpExtApplicationCustomizerProperties> {
private LOG_SOURCE = "SpExtApplicationCustomizer";
private _elementId = "SpExtApplicationCustomizer";
private _topPlaceholder: PlaceholderContent | undefined;
public async onInit(): Promise<void> {
try {
this.context.application.navigatedEvent.add(this, this._render);
} catch (err) {
console.error(`${this.LOG_SOURCE} (onInit) - ${err}`);
}
}
@override
public async onDispose(): Promise<void> {
try {
this.context.application.navigatedEvent.remove(this, this._render);
// eslint-disable-next-line @microsoft/spfx/pair-react-dom-render-unmount
ReactDOM.unmountComponentAtNode(this._getTopContainer());
} catch (err) {
console.error(`${this.LOG_SOURCE} (onDispose) ${err}`);
}
if (this._topPlaceholder && this._topPlaceholder.domElement) {
this._topPlaceholder.domElement.innerHTML = "";
}
}
private _getTopContainer(): HTMLElement {···
}
private async _render(): Promise<void> {···
}
main 0↓ 2↑ ⊗ 0 ⚠ 0 ⓘ 14 Live Share
Ln 8, Col 1 Spaces: 2 UTF-8 LF {} TypeScript Port : 5500 14 Spell


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package-lock.json
package.json
OUTLINE
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NPM SCRIPTS
CODETOUR
SpExtApplicationCustomizer.ts
SPExt > src > extensions > spExt > SpExtApplicationCustomizer.ts > SpExtApplicationCustomizer > onInit
26 public async onDispose(): Promise<void> {···
37 }
38
39 private _getTopContainer(): HTMLElement {
return document.getElementById(this._elementId);
41 }
42
43 private async _render(): Promise<void> {
44 try {
45 if (!this._topPlaceholder) {
46 this._topPlaceholder = this.context.placeholderProvider.tryCreateContent(
PlaceholderName.Top,
48 { onDispose: this.onDispose }
49);
50 }
51
52 //if a placeholder was retrieved, then we can work with that
53 if (this._topPlaceholder != undefined) {···
79 }
80 else {
81 console.error(`${this.LOG_SOURCE}(_render) Top Placeholder not available!`);
82 }
83 }
84 catch (err) {
85 console.error(`${this.LOG_SOURCE}(_render): ${err}`);
86 }
87 }
88 }
89
main 0↓ 2↑ 0 0 14 Live Share
Ln 17, Col 41 Spaces: 2 UTF-8 LF TypeScript Port : 5500 14 Spell


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SpExtApplicationCustomizer.ts
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.gitignore
.npmignore
.yo-rc.json
gulpfile.js
package-lock.json
package.json
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CODETOUR
SpExtApplicationCustomizer.ts
SPExt > src > extensions > spExt > SpExtApplicationCustomizer.ts > SpExtApplicationCustomizer > _getTopContainer
53 if (this._topPlaceholder != undefined) {
54 //get the nav container we want that would be inside of the SPFx placeholder
let container = this._getTopContainer();
56
57 //if the container was not found, go and create, then append to spfx top placeholder
58 if (container == undefined) {
59 //create nav div
container = document.createElement("DIV");
container.setAttribute("id", this._elementId);
62 //bind to top placeholder
63 this._topPlaceholder.domElement.appendChild(container);
64 }
65
66 //Set up header launcher
67 const items = [
{ title: "Test Item 1", description: "Lorem ipsum dolor sit amet, consectetur adipiscing elit", url: "https://bing.com" },
69 { title: "Test Item 2", description: "Lorem ipsum dolor sit amet, consectetur adipiscing elit", url: "https://bing.com" },
70 { title: "Test Item 3", description: "Lorem ipsum dolor sit amet, consectetur adipiscing elit", url: "https://bing.com" },
71 { title: "Test Item 4", description: "Lorem ipsum dolor sit amet, consectetur adipiscing elit", url: "https://bing.com" }
72];
73
const lib1Launcher: ILib1Launcher = new Lib1Launcher({
domElement: container,
76 items: items
77 } as ILib1LauncherProps);
78 lib1Launcher.launch();
79 }
80 else {
81 console.error(`${this.LOG_SOURCE}(_render) Top Placeholder not available!`);
main 0↓ 2↑ ⊗ 0 ⚠ 0 ⓘ 14 Live Share
Ln 41, Col 4 Spaces: 2 UTF-8 LF {} TypeScript Port : 5500 14 Spell


Lib1_1 Library Workbench
Lib1 Library Workbench
Demos - Home
sympjulie.sharepoint.com/sites/Demos
Apps for SharePoin...
Demos - Home
pnpjsteam - Home
SPFX Debug
pnpjs
N8D - Home
HTWOO
HTWOO-React
SolutionHealth - H...
https://sympweb.z1...
pnpjsteam - Site Co...
_api/$metadata
My list of items
Test Item 1
Lorem ipsum dolor sit amet, consectetur adipiscing elit
Launch
Test Item 2
Lorem ipsum dolor sit amet, consectetur adipiscing elit
Launch
Test Item 3
Lorem ipsum dolor sit amet, consectetur adipiscing elit
Launch
Test Item 4
Lorem ipsum dolor sit amet, consectetur adipiscing elit
Launch
D
Demos
Following
Share
Home
Documents
Pages
Apps for SharePoint
Site contents
Edit
New
Page details
Analytics
Published 6/1/2023
Edit
Elements
Console
Sources
Performance insights
Network
Performance
<!DOCTYPE html>
<html lang="en-US" dir="ltr">
<head></head>
<body class="ms-font-m"> == $0
Skip to main content
<script nonce="3f8uo1ap31j">__spfxMark('bodyStart')</script>

<script nonce="3f8uo1ap31j">__spfxMark('suiteNavRenderScriptStart')</script>
<script nonce="3f8uo1ap31j">__spfxMark('suiteNavScriptEnd')</script>
<script nonce="3f8uo1ap31j">__spfxMark('moduleLoaderScriptStart')</script>
<script nonce="3f8uo1ap31j">__spfxMark('moduleLoaderScriptEnd')</script>
<script nonce="3f8uo1ap31j">__spfxMark('postFlush')</script>
<script nonce="3f8uo1ap31j">__spfxMark('bodyEnd')</script>
<iframe src="https://substrate.office.com/search/api/v2/resources" style="display: none;"></iframe>
</body>
</html>
<!--EOC2(3ms)-->
Styles
Computed
Layout
Event Listeners
element.style {
}
body, html {
height: 100%;
}
body {
-webkit-overflow-scrolling: touch;
background-color: #FFFFFF;
color: #242424;
margin: 0;
overflow-x: hidden;
overflow-y: auto;
padding: 0;
}
body {
height: 100vh;
margin: 0;
padding: 0;
}
html, body {
margin: 0;
padding: 0;
height: 100%;
}
body {
display: block;
margin: 8px;
}
margin
border
padding
1056×975
html
body.ms-font-m
Console
Issues
Rendering
top
Filter
All levels
4 Issues: 1 3

Resources

NPM 5 changes to NPM Link

chevtek.io/npm-5-changes-to-npm-link/

NPM Workspaces

docs.npmjs.com/cli/v7/using-npm/workspaces

Self-Signed Cert for Testing

stackoverflow.com/questions/14267010/how-to-create-self-signed-ssl-certificate-for-test-purposes

Scan for slides & links:

